Reducing Emissions in the Laboratory*

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Abstract

Laboratory sustainability programs have been widely adopted, yet evidence for their effects rests on uncontrolled before-and-after comparisons. We provide the first causal evidence, using a randomized controlled trial with 109 research groups at the University of Zurich. The intervention, adapted from a program used by over 150 institutions, combines practice guidelines, personalized energy-use feedback, and certification. We find no detectable effect on energy use and rule out reductions larger than 14%, including those the program reports. Most groups already met the program's entry-level certification standard at baseline, suggesting such programs may largely recognize existing practices rather than change them.

JEL classification: C93, D83, Q41, Q54
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1 Introduction

Organizations worldwide have committed to ambitious emissions reduction targets. In Switzerland, the Climate and Innovation Act mandates net-zero greenhouse gas emissions by 2050. Organizations also seek to reduce energy use for its own sake, since doing so cuts costs, particularly as energy prices have risen. Yet within organizations we often observe an "energy paradox": energy-saving changes are not adopted despite being cost-effective (Jaffe and Stavins, 1994; Gillingham and Stock, 2018; Tiefenbeck et al., 2018). One reason is a classic principal-agent problem: the employees or other actors who could take action to save energy often do not bear the costs of failing to do so, while acting is costly to them. A second reason is informational: even a sufficiently motivated actor may not know how to effectively reduce emissions and energy use. Policies that increase incentives for organizational actors to reduce energy use and emissions, and convey information on how to do so effectively, are therefore key to achieving organizational emissions and energy reductions targets. Monetary incentives could in principle directly solve the principal-agent problem, but they are often infeasible within organizations, due to relatively inflexible pay structures and difficulties in monitoring individual energy use.

We focus on the setting of energy use by scientific research groups within a university. The principal-agent problem is particularly acute here: research groups do not bear the cost of their own energy use, and they are highly autonomous in their research activities. Furthermore, research groups may perceive energy-saving measures as risking harm to their research, even when they are intrinsically motivated to reduce their environmental impact. This setting is therefore well suited to policies that increase non-monetary incentives to act and provide information on energy-saving actions that do not compromise research quality. Laboratory sustainability programs are a prominent example: the Laboratory Efficiency Assessment Framework (LEAF) has been adopted by more than 150 institutions worldwide (UCL, 2026), and similar programs have spread quickly (My Green Lab, 2021). LEAF reports that participating groups reduced emissions by 2.9 tonnes of carbon dioxide equivalent (tCO_2e) per year, though this figure rests on uncontrolled before-and-after comparisons of laboratories that voluntarily participated in a pilot, rather than experimental variation (UCL, 2022). Since such programs require researchers' time and institutional resources, robust evidence on their effects matters for the many institutions adopting them.

In this paper, we provide the first causal evidence on the effects of a sustainability

policy intervention adapted from LEAF. We ask whether the intervention – designed to supply information and non-monetary incentives, without changing who bears the cost of energy use – causes research groups to reduce their energy use. To answer this question, we conduct a randomized controlled trial (RCT) at the University of Zurich, randomly assigning research groups in the science and medicine faculties to treatment or control. Treated groups receive the intervention, which has three components: sustainable practice guidelines, provided as a checklist of actions a group can take; feedback on the group's own energy use and potential savings, in the form of interactive calculators with energy-saving tips; and certification, recognizing groups that complete enough actions from the sustainable practice guidelines. We evaluate these components as a bundle: our estimates capture their joint effect relative to no intervention, rather than the effect of any single component.

The study was presented to research groups as a university-wide assessment of laboratory energy use. Our analysis sample comprises 109 groups that completed both a baseline and an endline visit, roughly two months apart, between September 2025 and March 2026. At each in-person visit, enumerators collected equipment-use data from which we calculate each group's energy use, using the same formulas as the calculators the intervention itself provides. We complement these data with the titles and abstracts of over 8,000 publications by groups in our sample, to which we apply a topic model. We find that research content predicts energy use beyond group size and institute, but leaves around 40% of the variation unexplained, suggesting that energy use is not fully determined by the research a group does. Given the frequent communication and collaboration between groups, we also examine spillovers to control groups, as well as effects on attitudes towards sustainability, including whether the intervention causes a backlash against sustainability policies.

We find no detectable effect of the intervention on calculated energy use, our primary outcome, either overall or in any pre-specified subgroup. Our estimates are precise enough to rule out large effects: the 95% confidence interval excludes reductions larger than 14% of baseline energy use, whereas the reduction reported in the LEAF pilot corresponds to a 64% reduction of mean baseline energy use in our sample. Energy use is also highly persistent: 79% of groups show no change between baseline and endline.

The absence of an effect does not appear to reflect a failure to reach groups: treatment increases reported awareness of energy and resource costs. Treated groups also reported taking more sustainable actions between visits, including energy-relevant ones,

raising the share eligible for certification. However, these reported changes are not reflected in groups' reported equipment use, from which our energy outcome is calculated. The scope for improvement was also limited: based on their baseline responses, 73% of treated groups would already have qualified for the entry-level certification standard, and half of the calculators contained no energy-saving tips. In addition, the two-month period between visits likely left little time to coordinate changes to established research practices. Consistent with this, over a third of treated groups reported not having discussed the sustainable practice guidelines at all between visits.

We find little evidence of spillovers to control groups, and no significant effects on most attitudes towards sustainability, including stated intentions to consider sustainability in decision-making and beliefs that sustainable practices can improve research. The exception is that treated groups are less likely to agree that they would be more motivated to act if their group directly faced the financial cost of energy use, though this result is significant at the 10% but not the 5% level after adjusting for multiple hypothesis testing. We also find no evidence of backlash: treated groups are no more likely to report that sustainability requirements risk slowing down research, and we detect no change in support for top-down initiatives.

We make three contributions. First, we contribute to the evidence on laboratory sustainability programs. Laboratories are among the most energy-intensive spaces a university operates, consuming several times as much energy per square metre as offices (NREL, 2008), yet the evidence supporting these programs has not kept pace with their spread: existing work documents adoption, self-reported change, and before-and-after comparisons without a control group, rather than causal estimates of the programs' effects on energy use (Franssen, Giannini and Johnson, 2022; Schell and Bruns, 2024; Padget et al., 2025; Di Marco et al., 2026). We provide causal evidence of the impact of an intervention closely based on LEAF, using the same energy measure as the pilot evaluation on which the 2.9 tCO_2e figure is based, and rule out reductions of that magnitude. We also provide causal estimates of the intervention's effects on researchers' attitudes and beliefs, where the existing evidence is correlational (Schell and Bruns, 2024).

Second, we test whether non-monetary incentives can substitute for a price when incentives are split inside an organization. Split incentives are well documented between landlords and tenants (Bird and Hernández, 2012; Gillingham, Harding and Rapson, 2012; Myers, 2020; Domenig et al., 2024), and between building owners and contractors (Blonz, 2023), but the same wedge operates inside organizations, where the peo-

ple choosing how to use energy are rarely the ones paying for it. Organizations cannot easily meter or price the consumption of individual groups, so they instead use non-monetary incentives such as recognition, targets, and certification (Handgraaf, Van Lidth de Jeude and Appelt, 2013; Staddon et al., 2016). Our setting offers a particularly clean case to investigate the effects of non-monetary incentives: research groups control energy-intensive equipment, face no price for using it, and are autonomous enough that the university cannot mandate practice, leaving certification as one of the few levers available. We find no detectable effect of certification, combined with information and feedback, on energy use in this setting, at least in the short run.

The closest evidence on reducing energy use in university laboratory settings comes from Gillis et al. (2022), who find that messages emphasizing personal efficacy and a non-monetary reward (a performance challenge) both reduced energy use from fume cupboards – equipment frequently used in laboratories to contain toxic gases and fumes – among US undergraduate chemistry classes. Several features of our setting may explain why we do not find a similar effect. First, the underlying behaviors differ in how easily they can be changed: their messages targeted closing fume cupboard sashes – a single, low-cost action with an immediate and visible effect – whereas the sustainable practice guidelines in our intervention cover 49 different actions. Second, we focus on established research groups with long-standing research practices, which are more deeply embedded and likely harder to shift within a short intervention period, unlike undergraduate students taking a class for the first time. The difference is unlikely to reflect a weaker intervention: alongside recognition through certification, our intervention also supplies personalized feedback on the group's own measured energy use. This suggests that the absence of an effect on energy use may be partly due to the nature of the behaviors and actors the intervention targets.

Third, we ask what personalized feedback achieves when the recipient does not bear the cost of energy consumption. Feedback interventions, including social comparison-based home energy reports, have been shown to reliably though modestly reduce residential energy use (Allcott, 2011; Allcott and Rogers, 2014; Jessoe and Rapson, 2014; Andor and Fels, 2018; Brülisauer et al., 2020). The mechanism is usually understood to run partly through the financial saving the household captures. It is less clear whether feedback remains effective when recipients lack this financial incentive. Myers and Souza (2020) find that social comparison-based home energy reports had no significant effect on heating demand among dormitory residents who do not pay their own energy bills.

Our results are consistent with that finding: feedback increased reported awareness of energy and resource costs, but did not reduce energy use, though the feedback element of our intervention does not include a social comparison.

The rest of the paper is structured as follows. Section 2 describes the setting and sustainability policy intervention. Section 3 describes the RCT. Section 4 explains our empirical strategy. Section 5 presents descriptive evidence on energy use and sustainable practices at baseline. Section 6 presents our results. Section 7 concludes.

2 Setting and Intervention

2.1 The university

We focus on a large public research university in Switzerland, the University of Zurich, which faces challenges common to many organizations of meeting emissions and energy reductions goals while relying on internal actors who may lack information and incentives to take action. The university has over 28,000 students and employs over 10,000 staff, making it the largest university in Switzerland. It comprises seven faculties and over 150 departments.

The university reported 23,000 tCO_2e of emissions in 2024, with a rising trend since 2020 (UZH, 2025). At the same time, the university has set a 2030 climate neutrality target, more demanding than the national obligation of net zero by 2050. The university reports flights as the largest source of emissions, followed by energy, then commuting. From 2018 to 2024, emissions from air travel decreased, while emissions from energy slightly increased (UZH, 2025). The university is also seeking to reduce its energy demand, which has significant financial costs. Around 60% of energy demand at the university is for electricity, while the remaining 40% is for heat. The university's purchased electricity mix is largely generated by renewables, and therefore relatively clean, at 22.23 gCO_2e per kWh.[1] In 2024, the university procured 62.9 GWh of electricity, equating to emissions of approximately 1,400 tCO_2e, thus still accounting for a significant share of total emissions (UZH, 2025). Our setting may therefore represent a lower bound for the energy-reduction effects of interventions of this kind: on a clean grid, a unit of en-

[1]Switzerland as a whole has a low carbon intensity of electricity. The carbon intensity of the average consumer electricity mix in Switzerland is 96.9 gCO_2e per kWh (Federal Office for the Environment, 2026). This figure has been revised over time; the energy use calculators (Section 2.2) used the value reported at the time of the intervention, 128 gCO_2e per kWh.

ergy saved translates into smaller emissions reductions, so if energy-saving behavior is partly motivated by concerns about emissions, that motivation is weaker here than in a higher-carbon setting.

We focus on the science and medicine faculties at the university. These are the second and fourth largest faculties, respectively. These faculties contain most of the laboratory space at the university, as well as a large amount of office space. They house the most energy-intensive research equipment in laboratories, such as ultra-low temperature freezers (ULT freezers) – freezers designed to preserve biological samples at extreme temperatures, typically operated between -70°C and -80°C – and fume cupboards. At one campus, laboratories account for 41% of total electricity consumption (Wimmer, 2022). In addition, the university estimated that ULT freezers alone account for roughly 5% of all electricity consumed (UZH, 2025). Laboratories also have a high use of consumables and water, and generate significant waste.

The science and medicine faculties are organized into departments or institutes spanning a wide range of fields, from forensic medicine to geography. Within them, researchers are typically organized into research groups, which operate with a high degree of autonomy over their research, though with frequent collaboration and communication across groups. A group typically comprises a principal investigator, one or two laboratory managers or technicians, and a rotating population of postdoctoral researchers, doctoral students, and master's students. Technicians hold most of the practical knowledge about equipment, but any group member may make decisions about how it is used: which freezer a sample goes into, whether a fume cupboard sash is lowered, whether a piece of equipment is left running overnight. Research practices are passed down within research groups, through formal onboarding and handover procedures, or more often informally, and can persist long after the original rationale for a particular practice is forgotten. Research groups do not bear the cost of their own energy use, which the university's estates services cover centrally. In addition, before the study, almost all research groups had no way of measuring their energy use; a small number had installed smart plugs to meter individual pieces of equipment themselves. The university held some data at the building level, but buildings house many research groups, and this data was not shared with groups.

Several sustainability initiatives are already in place at the university, varying in scale and origin: some are university-wide and driven by the central administration, while others are specific to a department or campus. Existing initiatives include a tax on air

travel in the science faculty, and collaboration among research groups to share sustainable laboratory techniques in the science and medicine faculties. The university also runs a centrally coordinated project toward the 2030 climate neutrality goal, which oversees the intervention described in this study. There are also several bottom-up initiatives and organizations focused on laboratory sustainability, as well as a sustainability committee in the science faculty. A pilot of the LEAF program had already reached 35 research groups before the study; we exclude these groups from the sample.

2.2 The sustainability policy intervention

We study a sustainability policy intervention targeted at research groups. The intervention aims to improve sustainability by providing information and incentives to researchers to take action to reduce the environmental impact of their research group. Our intervention is adapted from the Laboratory Efficiency Assessment Framework (LEAF) and the Sustainable Practices and Research Knowledge Hub (SPARKHub). LEAF is one of the most widely adopted laboratory sustainability certification programs, developed at University College London (UCL) and adopted by over 150 institutions worldwide (UCL, 2026), operating as a licence model where institutions pay to access the program and its materials. LEAF reports reducing participating research groups' emissions by 2.9 tCO_2e per year in a pilot evaluation of the program (UCL, 2022), equivalent to roughly 14 MWh of energy use per group.[2] SPARKHub builds on LEAF, but is free and open-access, and plans to offer guidelines for additional research activities beyond those covered by LEAF. It is still in development as of September 2026.

The intervention consists of three key components: sustainable practice guidelines, certification, and energy use calculators. They are described in detail below.

1. **Sustainable practice guidelines:** Research groups receive information on actions to take to reduce the environmental impact of their research, drawn from both LEAF's and SPARKHub's guidelines. These guidelines are based on best practices from sustainable research initiatives worldwide, and are designed to convey actions that do not compromise research quality. Information is provided as a checklist of actions, with a description of each action's impact and a box for groups to describe

[2] We convert emissions to energy use with the UK grid emissions factor of 207 gCO_2e per kWh for 2023 (Department for Energy Security and Net Zero, 2023), since the pilot took place in the UK. We compare the pilot to our results in energy rather than emissions terms because the carbon intensity of electricity is much lower in Switzerland (Section 2.1).

how they are addressing it. Actions are rated bronze, silver, or gold by the difficulty and time they require, and are grouped by research activity: general laboratory work, offices and travel, cold storage, chemistry, and fume cupboards (Appendix Table A.1 lists all actions). There are 49 actions in total: 16 bronze, 18 silver, and 15 gold.

2. **Energy use calculators:** Research groups have access to calculators that allow them to estimate the energy use and carbon emissions of ten common items of laboratory and office equipment: fridges, freezers, ULT freezers, glassware drying cabinets, microbiological safety cabinets (MSCs), cryostats, water baths, CO_2 incubators, block heaters, and IT equipment.[3] We use the same formulas as in the SPARKHub calculators to calculate energy use.[4] We use the carbon intensity of the average consumer electricity mix in Switzerland to calculate carbon emissions, following the approach in the SPARKHub and LEAF calculators. The calculators are interactive, allowing groups to update their inputs and see updated totals, reported by equipment type and for the group as a whole. Depending on a group's responses, the calculators also list tips for reducing unnecessary energy use and emissions, which groups can use to estimate potential savings from changing their practices.

3. **Certification:** Research groups are certified for the highest level they qualify for, following a nested threshold based on the checklist actions in the sustainable practice guidelines: bronze requires completing at least 70% of bronze actions; silver requires having achieved bronze and completing at least 70% of silver actions; and gold requires having achieved silver and completing at least 70% of gold actions. Certified groups receive a certificate, which they can print and submit as part of funding applications. In our study, certification is based on an audit by the enumerators; in the LEAF and SPARKHub programs, groups are instead typically audited

[3]Glassware drying cabinet are ventilated cabinets used to dry laboratory glassware, typically operated at 60°C. Microbiological safety cabinets are enclosed workspaces designed to protect the worker from pathogens. Cryostats are refrigerated chambers, typically kept at around -20°C, used to slice frozen tissue samples for microscopy. Water baths are heated containers of water to incubate samples. CO_2 incubators are chambers to maintain controlled conditions for the growth of cell cultures. Block heaters are devices to heat samples precisely and uniformly.

[4]The SPARKHub calculators are mostly identical to the LEAF calculators, with some minor updates. The calculators were developed from electrical measurements of devices. The calculators do not cover fume cupboards because the SPARKHub calculator formula for this equipment was not yet available at the time of the intervention. The formula became available afterwards, and we use it to calculate fume cupboard energy use at both baseline and endline.

by another participating group, and LEAF's thresholds are not rigidly fixed.

The intervention combines an information treatment with non-monetary incentives. The sustainable practice guidelines provide researchers with information on actions they can take within their group to reduce environmental impact. The energy use calculators provide feedback at a uniquely granular level, since energy use data had not previously been available at the group level at the university. Certification functions as a non-monetary incentive, recognizing groups for their sustainability efforts – though it may also carry a secondary, indirect monetary incentive if groups believe it improves their funding prospects.

3 Experimental Design

3.1 Recruitment

We identified 367 research groups in the science and medicine faculties from the faculty websites. We then excluded groups listed more than once, groups that had participated in a pilot of the LEAF program, and groups in which our enumerators worked, to avoid spillovers. We randomly assigned the remaining 313 groups to the treatment group (50%) and the control group (50%), stratified by faculty. We then assigned groups to enumerators, who each received a list of six groups (three treatment and three control). We randomized before recruitment for two reasons: enumerators delivered the intervention at the end of the baseline visit and therefore needed to know each group's assignment in advance, and assigning each enumerator equal numbers of treated and control groups balanced enumerators across arms.

Enumerators contacted the groups by email to invite them to participate, typically writing to the group's principal investigator listed on the faculty website, whom we asked to refer us to the responsible contact person if it was not them. The invitation presented the study as a university-wide assessment of laboratory energy and resource use, conducted as part of the university's climate neutrality initiative and announced in advance by the faculty deans. It asked groups to take part in an in-person survey on current practices of approximately two hours. Whenever a group declined, we gave the enumerator a replacement group from the same arm. In total, enumerators contacted 305 groups, of which 138 (45%) agreed to participate.[5] The most common reasons for

[5]The list of groups to contact changed during recruitment: we removed groups from institutes that de-

non-participation were non-response, unwillingness to participate, and being part of a larger group that was already participating.

3.2 Treatment

Treated groups received all three components of the intervention described in Section 2.2 as a bundle, while control groups received none of them. Our treatment effect is therefore the joint effect of the three components relative to control; the design does not allow us to separate the contribution of each. Enumerators delivered the guidelines and informed groups about certification at the end of the baseline visit. We emailed the calculators about one month later, and the certification audit took place at the endline visit (Table 1). We describe how each component was implemented below.

Sustainable practice guidelines. Treated groups went through the sustainable practice guidelines checklist with the enumerator at the end of the baseline visit. The checklist was in a workbook (Appendix Figures A.1 and A.2). For each action in the checklist, groups indicated whether they had already taken it, with the additional options of "I don't know" and "Not applicable" where relevant, and groups were encouraged to add notes or questions for the research team in a separate text box. After the visit, enumerators sent each group its completed checklist in a follow-up email and encouraged them to go through it with the rest of their research group.

Energy use calculators. One month after the baseline survey, we emailed treated groups energy use calculators, pre-filled with their baseline equipment responses. We built the calculators ourselves as workbooks, using the same formulas as the LEAF and SPARKHub calculators, which are also the formulas we use to construct our outcome. We encouraged groups to use the calculators to estimate potential energy savings. Appendix Figures A.3 and A.4 depict the calculator.

Certification. Enumerators told treated groups about the certification audit at the end of the baseline visit, and groups were reminded in two follow-up emails: the one from the enumerator with their completed guidelines, and the one from the research team with the energy use calculators. The audit took place at the endline visit, immediately

clined to participate at all, and we added several groups that we learned about in the process, which we randomized in the same way.

after the endline survey. Enumerators went through the sustainable practice guidelines with the group to confirm which actions it had taken, following the same procedure as at baseline, except that the responses now counted toward certification. In June 2026, groups received their certificate if they qualified. The certificate bears the SPARKHub logo (Appendix Figure A.5). Because the audit followed the endline survey, certification can affect our outcome only through groups anticipating it.

3.3 Data collection

Data collection took place during two in-person visits by enumerators (Table 1): a baseline visit from September to October 2025 and an endline visit from December 2025 to March 2026. The enumerators were PhD students from the science faculty and a neighboring university, as well as members of the research team. Each visit included a survey, administered as a workbook with one tab per section, which we sent groups ahead of the baseline visit so that they could review the questions in advance. At the baseline visit, enumerators introduced the survey as collecting data on current practices to better understand energy and resource use at the university. The enumerator typically completed the survey with the group's principal investigator, a laboratory technician, or a postdoctoral researcher. Sometimes enumerators completed the survey with several group members. We let groups choose who answered, because the person responsible for the equipment varied across groups and was unknown in advance. In treated groups, the same people also completed the sustainable practice guidelines, were told about the certification audit, and received the enumerators' and research team's emails. We arranged the endline visits with the same contacts as at baseline, but we did not record who answered the endline survey.

At baseline, groups were surveyed on their use of common laboratory equipment, such as fume cupboards, cold storage, and IT equipment. For each type of equipment, the questions cover the quantity and type of equipment and how the group uses it, since these determine its energy use (Appendix Figure A.6 shows the freezer questions as an example). Groups reported equipment in batches, that is, sets of pieces of the same type that share all characteristics. We compute the annual energy use of each batch of equipment with the formulas from the LEAF and SPARKHub calculators, which are derived from direct electricity measurements of the equipment under different conditions and with different settings. Our primary outcome is a group's total annual energy use across all equipment types and batches, measured in MWh.

The baseline survey also collected group characteristics: the faculty, institute, number of researchers, and rooms the group uses.[6] Because these were measured before treatment, we use them for heterogeneity analysis and as controls, and to construct measures of geographical proximity between groups. The survey also asked which other groups or researchers each group shares equipment and space with or communicates with, and how regularly. We use these data to construct networks of research groups and assess potential spillovers. We also asked for consent to collect and merge the survey data with publicly available data on the respondent's publications.

Enumerators conducted the endline survey during in-person visits from December 2025 to March 2026, with each group visited by the same enumerator as at baseline. The survey repeated the baseline questions on equipment use, from which we compute energy use at endline. To ease the burden on respondents, we pre-filled each group's baseline responses. To make sure groups reconsidered every entry, we added one question per equipment type asking whether door openings, days in use, or temperature had changed, which prompted groups to update all the relevant answers.

The endline survey also asked about attitudes towards the environmental impact of research at the university and towards top-down versus bottom-up approaches to changing research culture. These questions were optional as they pertained to the personal beliefs of the researcher. We use these responses to estimate the effects of the intervention on broader beliefs. We also asked treated groups about their engagement with the sustainable practice guidelines to measure how salient they were, and control groups whether they were aware of the sustainability program, to investigate spillovers.

Publications data. We also collected publication data to identify spillovers between research groups. In 104 of the 138 groups that participated at baseline (75%), the person who completed the baseline survey consented to this data collection. We searched the university's open access repository for all of their publications and found publications for 95 of these groups, yielding 8,014 publications published between 1985 and 2026. University researchers are required to publish all publications in this repository, with some legal exceptions. We use these data to find collaborations that run through individuals rather than through an established group-to-group equipment or space sharing relationship. We also use these data for descriptive analysis of the energy intensity of research.

[6] We ask groups for their faculty because the faculty websites, which we used to assign treatment, were sometimes outdated.

Table 1: Timeline of the intervention and data collection

Time	Type	Event	Groups	Delivery
Baseline visit (Sep–Oct 2025)	Data collection	Baseline survey	All groups	In person
Baseline visit (Sep–Oct 2025)	Intervention	Sustainable practice guidelines completed	Treatment	In person
Baseline visit (Sep–Oct 2025)	Intervention	Information about certification audit	Treatment	In person
Oct 2025	Intervention	Enumerator email: completed guidelines, certification reminder	Treatment	Email
Nov 2025	Intervention	Research team email: energy use calculators, certification reminder	Treatment	Email
Endline visit (Dec 2025–Mar 2026)	Data collection	Endline survey	All groups	In person
Endline visit (Dec 2025–Mar 2026)	Intervention	Certification audit	Treatment	In person
Jun 2026	Intervention	Certificates awarded	Treatment	Email

Note: This table shows the timeline of the intervention and data collection, in chronological order. Type indicates whether an event is part of the intervention or of data collection. Groups indicates which groups took part: all groups, or treated groups only. Dates give the range over which visits took place.

3.4 Sample

Of the 138 groups that participated in the baseline, 70 were assigned to treatment and 68 to control. Of these, 29 groups (21%) did not complete the endline visit: 14 treated groups (20%) and 15 control groups (22%) attrited between baseline and endline. Our analysis sample consists of the 109 groups – 56 treated and 53 control – with both baseline and endline data. Attrition does not differ significantly by treatment status (Appendix Table A.2). Groups that attrited had lower baseline fume cupboard energy use than retained groups, but this is unlikely to bias our treatment effect estimates: attrition is unrelated to treatment, and baseline characteristics remain balanced across arms in the analysis sample, apart from equipment sharing (Appendix Table A.3).

Table 2 describes the analysis sample at baseline. Of the 109 groups, 77 (71%) are in the science faculty, 27 (25%) in the medicine faculty, and 5 (5%) are joint between the science and medicine faculties. Because so few groups are joint, we combine them with the medicine groups and distinguish two categories in our analysis: Science (77 groups) and Medicine or Joint (32 groups). Groups vary considerably in size, with a mean of 12.4 researchers (SD 21.7) and a median of 8. Groups belong to 31 different institutes, of which 11 contribute only one group to the sample. 30 enumerators visited the groups, on average visiting 3.6 groups each.

Baseline annual energy use averages 22.0 MWh (median 13.3 MWh), or roughly 1.8 MWh per researcher – about 13 times the 0.14 MWh per year that a researcher using only standard IT equipment would consume.[7] Section 5.1 describes energy use in

[7]We calculate this using the IT equipment formulas from the energy use calculators, assuming one laptop and one external monitor (24–27 inches) on medium brightness, used during normal working hours (9

research in more detail. The most common equipment type is IT equipment (15 units per group on average), with all groups using at least one unit. Specialized equipment such as cryostats and glassware drying cabinets is less common (0.08 and 0.20 units per group on average, respectively), reflecting that these types of equipment are only used for certain types of research. In the analysis sample, 81 groups (74%) consented to the collection and merging of publication data, of which we match at least one publication to 74 groups (91%). On average, consenting groups have 87 matched publications (SD 160; median 29), reflecting a small number of groups with very large publication records.

hours a day, 255 days a year).

Table 2: Sample overview

	Mean	SD	Median	N
Sample composition				
Science	0.706			109
Medicine	0.248			109
Joint	0.046			109
Assigned to treatment	0.514			109
Number of researchers	12.38	21.70	8.0	109
Baseline annual energy use (MWh)				
Total	22.03	36.95	13.31	109
Baseline equipment (units per group)				
IT equipment	14.97	23.68	9.0	109
Fume cupboards	3.60	8.37	1.0	109
Fridges	3.56	5.51	2.0	109
Freezers	3.30	7.19	1.0	109
Block heaters	1.78	4.25	0.0	109
Water baths	1.26	2.95	0.0	109
CO_2 incubators	0.89	2.25	0.0	109
Microbiological safety cabinets	0.75	1.62	0.0	109
ULT freezers	0.65	1.08	0.0	109
Glassware drying cabinets	0.20	0.56	0.0	109
Cryostats	0.08	0.34	0.0	109
Total units	31.05	41.22	21.0	109
Publications (consenting groups only)				
At least one matched publication	0.914			81
Matched publications	87.25	160.21	29.0	81

Note: This table reports summary statistics for the analysis sample of 109 research groups with both baseline and endline data. For binary variables, the table reports the mean and number of observations (N); for all other variables, it reports the mean, standard deviation (SD), median, and N.

4 Empirical Strategy

This study was pre-registered in the AEA RCT Registry (AEARCTR-0016855), which specifies our primary outcome, pre-specified subgroups, and secondary outcomes. We describe our specifications below.

Primary outcome. Our primary outcome is calculated energy use in MWh: energy use estimated from survey responses on equipment use, entered into the energy use calculators (described in Section 2.2; these are the same calculators used as part of the treatment). The calculators transform survey responses about machine use and research practices into estimates of energy use, drawing on electrical measurements from comparable machines and calibrating these to reported settings and usage. We estimate the effect of the intervention using a difference-in-differences specification at the research-group-by-period level, where each research group is observed at two periods, a baseline period prior to treatment and an endline period following treatment:

$$Y_{it} = \beta \cdot Post_t \cdot Treated_i + \theta_t + \eta_i + \epsilon_{it} \tag{1}$$

where Y_{it} denotes calculated energy use for research group i at period t. $Post_t$ is a dummy for the post-treatment (endline) period. $Treated_i$ is a dummy for treatment assignment, θ_t denotes period fixed effects, and η_i denotes research group fixed effects. We estimate this specification both in levels and with a log transformation of calculated energy use. Our hypothesis is that $\beta < 0$, that is, that the intervention reduces energy use.

Baseline characteristics are balanced across arms, with one exception: control groups are more likely to report sharing equipment with another group than treatment groups (Appendix Table A.3). Because our difference-in-differences specification includes research group fixed effects, any time-invariant differences between treatment and control groups are absorbed.

Spillovers. Research groups within the university often talk to each other, collaborate on research projects, and even share equipment and laboratory or office space, which may lead to spillovers between treated research groups and control research groups. We therefore extend Equation 1 to account for possible spillovers between research groups, estimating the following specification, again at the research-group-by-period level:

$$Y_{it} = \beta \cdot Post_t \cdot Treated_i + \gamma \cdot Post_t \cdot S_i + \theta_t + \eta_i + \epsilon_{it} \tag{2}$$

where $S_i = \sum_{j \neq i} w_{ij} Treated_j$ is exposure to other treated research groups for research group i, with w_{ij} denoting weights based on network proximity between research groups i and j. We normalize weights so that $S_i \in [0, 1]$. We construct S_i using pre-treatment

measures of network proximity, so that exposure is not affected by treatment. We consider several measures of network proximity between research groups: (i) geographical distance between laboratory or office spaces; (ii) laboratory or office space sharing; (iii) equipment sharing; (iv) communication; (v) research collaboration.[8]

Standard errors and robustness. In both specifications above, standard errors are clustered at the research group level. As robustness checks, we report p-values obtained using wild cluster bootstrap procedures and randomization inference. We also estimate the specifications with faculty-by-period, institute-by-period, and enumerator-by-period fixed effects.

Secondary outcomes: Awareness and attitudes. We are also interested in the effects of the intervention on researchers' awareness of the energy and resource costs of their group's activities, as well as on climate attitudes and attitudes towards top-down versus bottom-up initiatives. For outcomes where we only have post-treatment data, we estimate the following specification at the research group level:

$$Y_i = \beta_0 + \beta_1 Treated_i + \beta_2 X_i + \epsilon_i \tag{3}$$

where Y_i is the outcome of interest and X_i are baseline covariates: an indicator for the science faculty, log group size, and log baseline calculated energy use. Standard errors are clustered at the enumerator level.

Heterogeneity. For both our primary energy use outcome and the awareness and attitudes outcomes described above, we examine heterogeneity by faculty (Science versus Medicine or Joint), baseline energy use, research group type ("dry lab", groups with only IT equipment, versus "wet lab", groups with other types of laboratory equipment), and group size. For baseline energy use and group size, we split groups into those above or below the sample median. We estimate the specifications separately for each subgroup.

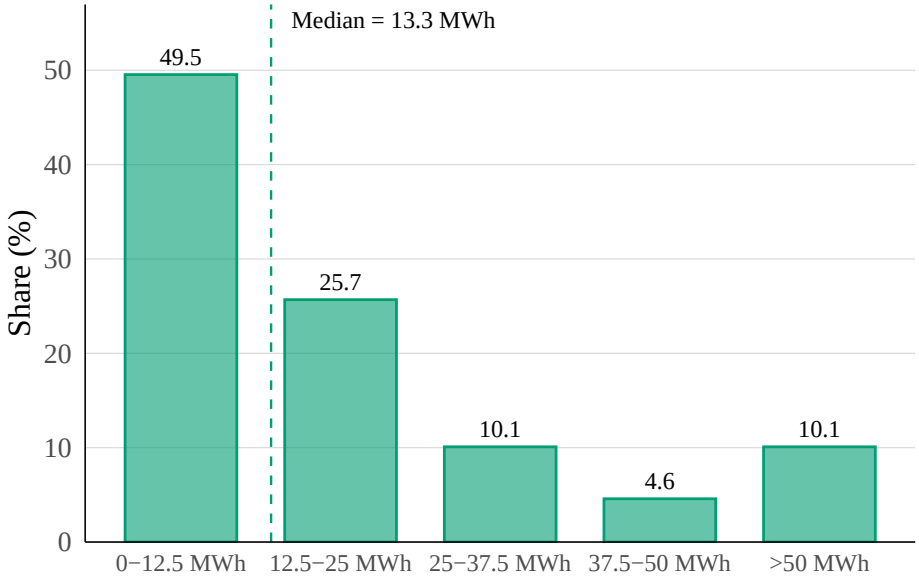
[8]The specification with the research collaboration measure is estimated on the subsample of research groups that consented to publication data collection and linkage; see Section 3.3 for details.

5 Descriptive Evidence

5.1 Energy use at baseline

We first document heterogeneity in research groups' energy use at baseline. Most groups have low energy consumption, with several large outliers. Figure 1 shows that median energy consumption is 13.3 MWh, while approximately 10% of groups report energy consumption over 50 MWh. This right-skewed distribution motivates our log specification. We also see heterogeneity by faculty: median energy consumption for groups in the science faculty is 9.4 MWh, compared to 19.0 MWh for groups in the medicine faculty or part of both faculties (Appendix Figure A.7). Only 6.5% of groups in the science faculty have annual energy use of more than 50 MWh, versus 18.8% of groups in the medicine or in both faculties. These differences motivate the heterogeneity analysis by faculty.

Figure 1: Annual energy use at baseline

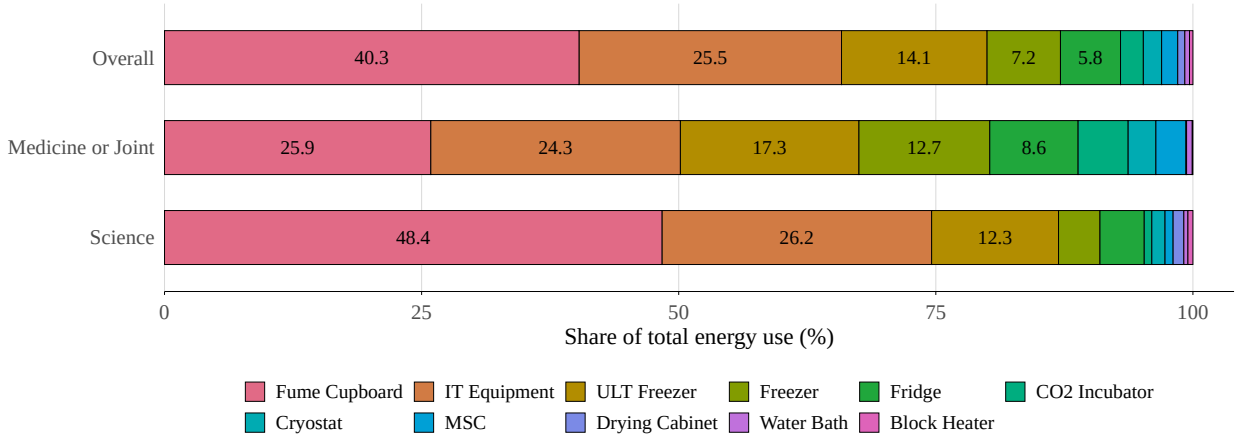


Note: This figure shows the distribution of annual energy use at baseline, in MWh. The sample includes the 109 groups in the analysis sample. The dashed line indicates the median.

Figure 2 illustrates that several types of equipment are responsible for a large share of energy use. Fume cupboards account for around 40% of overall energy use. IT equipment represents around one quarter of energy use. Cold storage – ULT freezers, freezers, and fridges – together make up another quarter of energy use. To some extent,

this reflects the number of units of equipment types: on average, groups have 15 units of IT equipment, while the average number of cryostats is 0.08. Appendix Table A.4 shows that the equipment with the highest mean energy use per unit is ULT freezers (4.8 MWh), followed by cryostats (4.7 MWh), and fume cupboards (2.5 MWh). We also observe heterogeneity by faculty: fume cupboards make up a larger share of total energy use for groups in the science faculty than groups in the medicine faculty or part of both faculties, while cold storage, CO_2 incubators, and MSCs account for a higher share of energy use for groups in the medicine faculty or in both faculties. This likely reflects the differences in the research being done in each faculty, requiring different types of equipment.

Figure 2: Energy use by equipment type



Note: This figure shows the share of total baseline annual energy use accounted for by each equipment type, overall and by faculty. The sample includes the 109 groups in the analysis sample. Drying Cabinet refers to glassware drying cabinet.

We next examine whether the content of a group's research – not just its size or institute – predicts its energy use. For the groups with publications data, we fit a topic model to the titles and abstracts of their publications in the university's repository. The model yields four research topics: immunology and virology, two particle physics topics (collider and flavor physics), and a broad residual category; we summarize each group's research by its average topic shares across its publications (Appendix Section B.1). We then compare how well two LASSO specifications predict log annual energy use: one with administrative characteristics only (log group size and institute fixed effects), and one that adds research topic shares.

Table B.3 shows that adding research topics increases the cross-validated R^2 from 0.49 to 0.61, indicating that research content carries information about energy use beyond group size and institute. Groups working on immunology and virology use more energy than groups doing general research: OLS estimates show that a 10 percentage point higher share of publications in immunology and virology is associated with roughly 31% higher energy use, and 43% when comparing only groups within the same institute, so the association is not driven by differences between institutes (Appendix Table B.2). This is consistent with immunology and virology requiring energy-intensive equipment such as cryostats for tissue processing and cold storage for sample preservation. The two particle physics topics are not distinguishable from general research. These results may partly reflect that our measure of energy use only captures a limited portion of the total energy use resulting from a group's research. In particular, we cannot measure any collider-related energy use among research groups (either within the university or externally). Group size also strongly predicts energy use. However, around 40% of the variation in energy use remains unexplained by research topics or group characteristics, suggesting that energy use is not intrinsically tied to a group's research output, and indicating that there may be scope for groups to reduce their energy use without compromising research quality.

5.2 Sustainable practices at baseline

The intervention revealed that many groups in our sample had already implemented sustainable practices in their research. Half of the calculators received by treated groups contained no energy-saving tips (Appendix Table A.5). In addition, no tips were displayed to any groups about IT equipment, reflecting sustainable existing practices around IT equipment. However, the tips displayed regarding cold storage suggest scope for improvement in energy efficiency. Approximately 60% of groups using a freezer received the tip to de-ice their freezer, while 46% of groups using a fridge received the tip to reduce door openings. 41% of groups using a ULT freezer were recommended to warm the ULT up to -70°C. In addition, a large majority of groups using CO_2 incubators received the tip to reduce usage to 255 days per year, suggesting that the default for this unit is "always on" across the organization. We expected that any reductions in energy use would be more likely to result from equipment for which groups received the energy-saving tips.

The responses to the sustainable practice guidelines checklist also provide evidence of

strong sustainable research practices in place at baseline. Appendix Table A.1 shows that for most checklist questions a high share of groups reported having taken that action at baseline. All treated groups using fume cupboards reported closing sashes when not in use, while seven other actions were reported by more than 90% of groups, including the regular maintenance of equipment, considering the repair and reuse of existing equipment before purchasing new equipment, and appropriate use of purified water sources. Based on these answers, 73% of treated groups would have been eligible for bronze certification at baseline, 44% for silver certification, and 35% for gold.[9] Combined with the fact that half of the treated groups received no tips based on their reported baseline equipment use, this suggests limited scope for the intervention to induce further sustainability improvements in our sample of research groups. The reason for such strong existing sustainability practices may be the initiatives already in place at the university at the time of the intervention.

6 Results

6.1 Effects on energy use

Table 3 presents results for energy use, our primary outcome. Across both levels and log specifications, we find no statistically significant effect of the intervention on equipment energy use. This absence of an effect holds in our heterogeneity analysis: Appendix Tables A.9 and A.10 show no significant treatment effects for any subgroup. The coefficient for the medicine and joint faculty subsample is larger in magnitude than for the science subsample, though neither reaches significance.[10] Nor do we find evidence that the intervention led to larger reductions among more energy-intensive groups, larger groups, or wet lab groups, contrary to our hypotheses. Disaggregating by equipment type yields the same pattern: Appendix Tables A.6 and A.7 show no significant effects for any individual equipment category.

[9] Certification levels are nested: gold requires silver, and silver requires bronze, so groups eligible at a higher level are also counted at lower levels.

[10] For the medicine and joint subsample, the levels estimate is significant under randomization inference ($p = 0.019$), but not with clustered standard errors, the wild cluster bootstrap, nor in the log specification.

Table 3: Treatment effects on energy use

	Levels	Log
	(1)	(2)
Treated × Post	-0.067	-0.056
	(0.102)	(0.051)
WCB p-value	0.540	0.200
RI p-value	0.507	0.176
Number of observations	218	218
Baseline mean	22.026	2.026

Note: This table reports the coefficient on Treated × Post from a difference-in-differences regression of energy use on treatment status, in levels (MWh) and logs, with research group and period fixed effects, estimated using Equation 1. The sample includes the 109 groups in the analysis sample, observed at baseline and endline (N = 218). Baseline mean reports the baseline mean of the outcome in column (1), and the baseline mean of the log outcome in column (2). Standard errors are clustered at the research group level and reported in parentheses. WCB and RI report wild cluster bootstrap and randomization inference p-values, respectively. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

To assess the magnitude of our estimates, we describe them relative to baseline energy use. In our log specification, the point estimate corresponds to a 5.4% reduction in energy use, with the 95% confidence interval ranging from a 14.4% reduction to a 4.5% increase.[11] Although we cannot rule out modest effects, our estimates are precise enough to rule out large ones. The reduction reported by the LEAF pilot, roughly 14 MWh per group (Section 2.2), corresponds to 64% of mean baseline energy use in our sample – more than four times the 14.4% reduction at the lower bound of our confidence interval.

Energy use is also highly persistent at the research group level: 79% of groups experienced zero change in equipment energy use between baseline and endline. The absence of any detectable subgroup or equipment-type heterogeneity suggests that the null holds across different kinds of research groups, or for equipment where energy-saving measures may plausibly be easier to implement. Together, these patterns are consistent with equipment energy use being shaped largely by the scale and nature of a group's research activities and by entrenched research practices, which are slow-moving and not easily shifted by an information- and feedback-based intervention. Changing these practices

[11] We convert log coefficients to percentage changes as $e^{\beta} - 1$. Our levels specification gives an even smaller effect: a reduction of 0.3% of mean baseline energy use, with a 95% confidence interval ranging from a 1.2% reduction to a 0.6% increase.

is particularly demanding in laboratory settings, where it requires agreement among group members and adjustments to established protocols, including health and safety procedures.

Our results stand in sharp contrast to the UCL (2022) pilot evaluation. The pilot's reported reduction exceeds the median baseline energy use in our sample, implying that the average participating group eliminated more energy use than a typical group in our sample consumes. Three differences may explain this gap. First, the pilot relied on a before-and-after comparison among groups that self-selected into the program, with no control group, whereas our estimate is based on random assignment. Second, pilot groups self-reported their savings, verified by audit processes that varied by institution, whereas in our experiment, enumerators collected and validated the data directly. Third, for a reduction of this size to be feasible, pilot groups likely had substantially higher baseline energy use than groups in our sample. We requested further detail on the pilot, but group-level data are no longer available, which limits how far we can assess this comparison.

6.2 Mechanisms

The absence of an effect on energy use does not appear to reflect a failure of the intervention to reach groups. First, the intervention increased researchers' awareness: Appendix Table A.11 shows a significant positive effect on reported awareness of the energy or resource costs of the research group's activities, suggesting that the intervention was effective in its intended information channel.[12] Second, treated groups reported adopting more sustainable practices. The share of treated groups meeting each certification threshold increased between baseline and endline: from 73% to 85% for bronze, from 44% to 60% for silver, and from 35% to 50% for gold (Appendix Figure A.9). Because only treated groups completed the checklist, and endline responses counted toward certification, this comparison has no control group and may partly reflect changes in reporting incentives; we therefore interpret it as suggestive evidence of treatment uptake.

Many of the practices in the sustainable practice guidelines are not reflected in our measure of energy use. Of the 49 actions, only 6 map directly onto inputs of our equipment energy use measure – such as cold storage management, fume cupboard sash

[12]The effect is significant at the 5% level with standard errors clustered at the enumerator level and using wild cluster bootstrap inference ($p = 0.026$), but not using randomization inference ($p = 0.113$); we therefore interpret it with some caution.

closure, and equipment switch-off – and a further 8 relate to them only indirectly. The majority of actions target sustainability in other areas including travel, water use, waste, and chemicals (Appendix Table A.1). Most actions were already reported as taken at baseline (74% of group–action pairs), leaving limited room for change. Between baseline and endline, treated groups newly reported 1.2 actions on average, of which 0.4 were directly or indirectly energy-relevant. The null effect is therefore not explained by groups prioritizing the non-energy-relevant actions: groups newly reported 19.0% of the energy-relevant actions not yet taken at baseline, compared with 13.5% of other actions (difference not statistically significant; Appendix Table A.12). However, none of the 8 instances in which a treated group newly reported an action that maps directly onto our energy use measure is matched by a corresponding change in the endline equipment survey. Since both the checklist and the endline equipment survey were prefilled with baseline responses, anchoring alone cannot explain this discrepancy. It may instead reflect over-reporting in the checklist, which counted toward certification, or the limited granularity of the equipment survey: groups report equipment use by selecting from a limited set of drop-down options, so small changes might not be captured.

Two further factors limited the scope for energy reductions. First, the energy use calculators identified few opportunities for savings: half of the treated groups received no energy-saving tips in their calculators (Section 5.2). Second, treated groups report limited engagement with the sustainable practice guidelines checklist over the intervention period. 34% reported that they had not discussed the checklist at all within their research group since the baseline visit, while 34% had only discussed the checklist a little (Appendix Figure A.8). This lack of engagement may reflect limited scope for further improvement, as well as the short period between visits, which gave groups few opportunities to discuss and coordinate changes.

Taken together, these results suggest that the intervention shifted awareness and reported practices, including energy-relevant ones, but that these changes did not translate into measurable changes in how groups use their equipment. The null effect on energy use is therefore informative: an intervention combining information, feedback, and certification can change what research groups know and what they report, but it is unlikely to deliver large energy savings, at least in the short run, in a setting where sustainable practices are already widespread. With 73% of treated groups already meeting the entry-level bronze certification standard at baseline, such programs may largely recognize existing practices rather than change them. The gap between reported practices and equipment

use also raises a concern for certification-based programs more generally: even with an in-person audit, reported practices may overstate actual changes, which suggests caution in evaluating such programs on the basis of reported actions or self-reported savings.

6.3 Spillovers

In our setting of research groups within a single university, we expect a high degree of cross-group interaction and communication. In many institutes and faculties, the divisions between groups are not clearly delineated, and researchers often work closely together. Ties across study arms are of particular interest, since they could expose control groups to the intervention. Consistent with connections across study arms, 35% of groups in our sample reported sharing equipment with a group in the opposite study arm, 28% reported sharing laboratory or office space with a group in the opposite arm, and among groups with publications data, 44% had co-authored at least one paper with a group in the opposite arm. These connections matter for two reasons. First, they represent a potential threat to the validity of a difference-in-differences approach that does not account for the presence of spillovers. Second, spillovers themselves could be policy-relevant – network ties between groups could enable the intervention to propagate beyond directly treated groups, amplifying the impact.

However, we find little evidence of spillovers of the intervention from treatment to control groups. In the endline survey, only one third of control groups reported that they had heard of any laboratory sustainability programs such as LEAF or SPARKHub, and fewer than 20% of those groups had first heard of such a program within the previous two months, suggesting that control groups remained unaware of the intervention throughout the study period. This is corroborated by the results of our specification accounting for spillovers: Table 4 shows that the small and insignificant treatment effect is robust to controlling for five different measures of proximity to treated groups, and the interaction between exposure and post-treatment is insignificant across four of the five measures. The exception is the collaboration-proximity measure, where the coefficient is negative and significant at the 10% level. With five proximity measures, one coefficient significant at the 10% level is close to what we would expect by chance, and this specification is estimated on the smaller subsample of groups with publications data. Taken together, control groups' lack of awareness of laboratory sustainability programs and the regression results both point to a lack of meaningful spillovers from treated to control groups.

Table 4: Treatment effects on energy use with spillovers

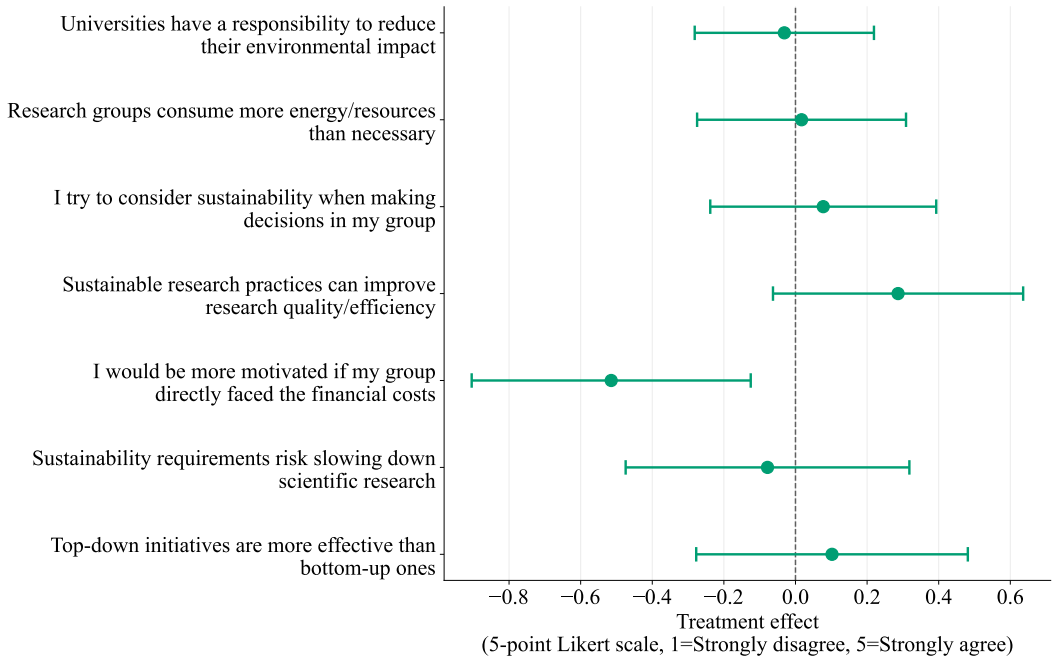
		Proximity measures				
	Baseline	Distance	Space	Equipment	Communication	Collaboration
	(1)	(2)	(3)	(4)	(5)	(6)
Treated × Post	-0.067	-0.062	-0.063	-0.089	-0.063	0.026
	(0.102)	(0.098)	(0.102)	(0.106)	(0.101)	(0.096)
Exposure × Post		-2.777	-0.092	-0.174	-0.128	-0.136*
		(1.882)	(0.102)	(0.127)	(0.100)	(0.078)
Number of observations	218	212	218	218	218	162
Baseline mean	22.026	22.213	22.026	22.026	22.026	24.210

Note: This table reports the coefficients on Treated × Post and Exposure × Post from a difference-in-differences regression with research group and period fixed effects, estimated using Equation 2. Column (1) reproduces the baseline specification from Table 3 without a proximity control. Columns (2) to (6) represent different proximity measures used to proxy for exposure to treated groups. The sample includes the groups in the analysis sample, but varies by column due to missing values in the proximity measures. Standard errors are clustered at the research group level and reported in parentheses. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

6.4 Attitudes

Before turning to treatment effects on attitudes, we examine attitudes among control groups to understand beliefs in the absence of the intervention. Appendix Figure A.10 shows high support for sustainability in universities: over 89% agreed or strongly agreed with the statement that universities have a responsibility to reduce their environmental impact, and 78% with the statement that they try to consider sustainability when making decisions in their group. These high levels of agreement may indicate ceiling effects for these statements. We also see mixed support for sustainability policies in research: 28% agreed that sustainable research practices can improve research quality or efficiency, while 26% agreed that sustainability requirements risk slowing down scientific research. 35% agreed that research groups consume more energy or resources than necessary, suggesting some acknowledgment of the need for policies like our intervention. We also observe that around two thirds agree that they would be more motivated if their group directly faced the financial costs of energy use, suggesting that monetary incentives may be an important motivator. Finally, we see mixed opinions on support for top-down versus bottom-up initiatives, with 46% agreeing that top-down initiatives are more effective.

Figure 3: Treatment effects on attitudes



Note: This figure shows the coefficient on treatment from separate regressions of each attitude outcome on treatment status, controlling for log group size, log baseline energy use, and faculty, estimated using Equation 3. The sample comprises 93 groups who answered the awareness and attitudes questions at endline. Outcomes are measured on a five-point Likert scale, from 1 (strongly disagree) to 5 (strongly agree). Standard errors are clustered at the enumerator level. Bars show 95% confidence intervals.

Figure 3 presents treatment effects on attitudes. We find no significant effects on beliefs about universities' environmental responsibility or on intentions to consider sustainability in decision-making – consistent with the high baseline agreement on these statements among control groups. We also find no significant effects on the belief that research groups consume more energy or resources than necessary, or that sustainable research practices can improve research quality or efficiency; baseline agreement here was moderate, so these insignificant results are not simply a ceiling effect. The only coefficient significant before adjusting for multiple hypothesis testing is a negative effect of -0.51 on the five-point Likert scale on agreeing that they would be more motivated to take energy-saving action if their group directly faced the financial costs of energy use.[13] Since treated groups were told they could receive certification, this may reflect the intervention making non-monetary rewards more salient relative to financial ones, even

[13]This result (raw $p = 0.010$) is significant at the 10% but not the 5% level after adjusting for multiple hypothesis testing across the 7 attitude questions (Benjamini-Hochberg $q = 0.067$; Holm $p = 0.067$).

if this did not translate into energy reductions; it could also reflect experimenter demand effects. We also find no evidence of a backlash to the intervention: treated groups were no more likely to agree that sustainability requirements risk slowing down research, and no less likely to agree that top-down initiatives are more effective than bottom-up ones, than control groups.

7 Conclusion

We study whether a sustainability policy intervention that supplies information and non-monetary incentives, without changing who bears the cost of energy use, reduces energy use among university research groups. In a randomized controlled trial with 109 research groups, an intervention adapted from LEAF – combining sustainable practice guidelines, energy-use feedback, and certification – has no detectable effect on calculated energy use. Our estimates rule out reductions larger than 14% of baseline energy use, including those reported in LEAF's own uncontrolled pilot evaluation.

The absence of an effect does not appear to reflect a failure to reach groups: treated groups report greater awareness of energy and resource costs and report adopting new sustainable practices. However, these reported changes are not reflected in groups' equipment use, and most groups already met the entry-level certification standard before the intervention. Taken together, our results suggest that certification-based programs may largely recognize existing practices rather than change them, and that evaluations based on reported actions or self-reported savings may overstate their effects.

These results should be interpreted in light of two limitations. First, the two-month period between baseline and endline may not have been long enough for new practices to become embedded in research routines. Second, our outcome is calculated from reported equipment use, which may not capture all margins on which groups changed their practices, and covers only part of groups' environmental impact.

Together, our results suggest that information and non-monetary incentives are a weak substitute for prices when the people who use energy do not bear its cost. Treated groups were also less likely to report that they would be more motivated if their group faced the financial costs of energy use, though this result does not survive adjustment for multiple hypothesis testing at the 5% level; whether monetary incentives would resolve the split-incentive problem in this setting remains an open question.

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A Appendix Figures and Tables

A.1 Tables

Table A.1: Checklist actions

Question	Category	Energy-relevant	BL Share (%)
Bronze Checklist			
A full assessment of laboratory equipment is completed to determine which items can be switched off and there are systems in place to maintain power management of equipment.	General Lab	All equipment	77
Purified water sources are used appropriately based on an assessment of the purity required for specific tasks.	General Lab		>90
In-House Production is utilised to reduce wastage and save resources.	General Lab		81
Recycle: Waste streams are properly separated to/and allow recycling with laboratories	General Lab		>90
Onboarding Training: All new laboratory members receive a sustainability induction.	General Lab		53
Prior to the purchase of new equipment, the group considers repair, purchasing used equipment, or using existing equipment	General Lab		>90
Paper and printing is minimised	Offices & Travel		84
Energy-Saving Modes: All electronic devices and computer screens, are configured to use energy-saving settings where compatible with accessibility	Offices & Travel	IT (partial)	73
Commuting: Employees are encouraged to commute by bicycle or public transportation. Carpooling is recommended where public transport is not an option.	Offices & Travel		78
ULT freezers are managed and maintained to optimise energy efficiency performace.	Cold Storage	ULT freezer	77
Swap water-jet aspirators for vacuum pumps	Chemistry		–
Solvent Emission Control: Waste containers for solvents are equipped with closed funnels or lids to minimize evaporation and volatile emissions into the (laboratory) environment.	Chemistry		>90
Waste canisters: reuse empty solvent jerrycans as labeled waste cans	Chemistry		62
Reuse cartridges for automated flash column chromatography systems up to 5x (reverse phase up to 100x) as suggested by the supplier.	Chemistry		–
Green Engineering Principles: 12 Principles of Green Engineering as well as the Sandestin Declaration: 9 Principles of Green Engineering are known to all researchers, displayed in the laboratory, and applied where scientifically feasible to minimize the environmental impact of chemical processes and engineering.	Chemistry		–
All fume cupboard sashes are kept closed when not in use	Fume Cupboards	Fume cupboard	100
Silver Checklist			
Equipment operation is optimised for energy efficiency.	General Lab	All equipment (partial)	79
Equipment maintenance and longevity: Laboratory equipment is regularly maintained and serviced to extend operational lifespan and ensure energy efficiency	General Lab	ULT, freezer (partial)	>90
Laboratory water consumption is minimised through equipment operation, maintenace and water-saving measures.	General Lab		82
Reuse: The lab has reviewed its use of consumables and identified those which may be reused.	General Lab		81
Experimental design is optimised (for material efficiency). Sample numbers, volumes and materials consumed are minimised. Workflows are organized to maximize material efficiency.	General Lab		84
Engagement of sustainability is managed in the laboratory through team or coordination.	General Lab		49
For the purchase of any consumables, those with the lowest environmental impact are selected.	General Lab		19
When purchasing, suppliers are engaged for low-carbon shipping options (take-back schemes, reduced packaging, and ambient temperature delivery) and these are implemented.	General Lab		41
Accessibility of Venues: Event venues are selected based on good access via public transportation. If this is not feasible, organized shuttle services or shared transport options are provided. Directions to the venue include information on environmentally friendly travel options.	Offices & Travel		84
Digital Communication: Unnecessary newsletters and mailing lists are unsubscribed from. Large files are shared via secure links rather than email attachments to reduce data transfer loads.	Offices & Travel		69
Reduced Travel Frequency: Participation in recurring conferences is optimized by reducing the frequency of attendance (e.g., every second event) or by joining regional conference hubs to avoid long-distance travel.	Offices & Travel		80
Samples and reagents are stored at appropriate temperatures for preservation and to minimise the energy demand of cooling devices.	Cold Storage	ULT, freezer (partial)	80
Switch to safer and renewable solvents (and eluents for chromatography) by default.	Chemistry		50
Rightsize reactions: run reactions on the smallest scale possible, run pilot/DoE and literature-informed scouting before scale-up; optimize for the right amount of solvent during extraction, apply the 12 Principles (catalysis, safer solvents, atom economy) and use biobased building blocks where possible.	Chemistry		>90
Replace oil pumps with scroll or screw pumps	Chemistry		43
Consider improving your high-pressure liquid chromatography (HPLC) purification by using HPLC columns with smaller inner diameter to reduce solvent consumption and waste creation.	Chemistry		–
Rotary evaporators: Ensure water baths have a cover. For instance, use plastic water bath balls for rotary evaporators to cover the open water. They prevent heat loss, evaporation (up to 90%), reduce oxygen input, reduce odors and levels of algae growth in the water. Furthermore avoid running the rotary evaporator overnight, unless required.	Chemistry	Water bath	50
Any permanently stored equipment in the fume cupboard is mounted to permit a flow of air underneath, possibly through supporting tiles	Fume Cupboards	Fume cupboard	36
Gold Checklist			
For the purchase of new equipment, relevant sustainability and performance criteria are utilised in technical specifications	General Lab	All equipment (partial)	53
Equipment is shared (where feasible) using systems to share equipment locally, across the organisation and externally (as appropriate).	General Lab	All equipment (partial)	>90
Autoclave and glasswasher operations are optimised in consideration of water and energy efficency.	General Lab		81
Harmful or environmentally impactful chemicals and reagents are reviewed for substition for more ecological options.	General Lab		53
Centralized Ordering: A designated person is responsible for lab orders, and orders are conducted on a weekly basis (not daily).	General Lab		88
Reproducibility of experiments is critically considered regularly.	General Lab		88
Sustainable Event Organization: When hosting events, environmental aspects are prioritized.	Offices & Travel		54
Virtual and Hybrid Meetings: Online or hybrid conferencing formats are considered to reduce travel emissions while maintaining collaboration and accessibility.	Offices & Travel		90

Question	Category	Energy-relevant	BL Share (%)
Avoiding Air Travel: Business flights are avoided whenever possible. When air travel is unavoidable, high-quality carbon offset programs are used to compensate for emissions.	Offices & Travel		84
Sample storage and freezer management practices optimise ultilisation of freezer storage space.	Cold Storage	ULT, freezer (partial)	–
End single-pass water cooling:Single pass cooling should be avoided to preserve water resources.	Chemistry		–
Replace water cooled condensers with air-cooled or Vigreux condensers (air-cooled condensers work the best for solvents with boiling points $> 50\,^{\circ}$C); for low-b.p. solvents, use closed-loop recirculators/thermostats. Furthermore add paddle-wheel flow meters on any recirculating loops.			
Replace oil baths with metal heating blocks.	Chemistry	Heat block, bath (partial)	–
Implement an electronic lab journal.	Chemistry		–
Consider the optimization of your purification process: an extraction, distillation, or recrystallization process can be faster, less expensive, and less solvent consuming than column chromatography.	Chemistry		–
When fume cupboards are due for annual inspection, owners ensure that the rates of extraction are not increased unnecessarily.	Fume Cupboards	Fume cupboard	52

Notes: This table lists the actions in the sustainable practice guidelines checklist, grouped by certification level and research activity (Category). BL Share reports the share of treated groups that reported having taken each action at baseline, among groups answering "Yes", "No" or "I don't know"; groups answering "Not applicable" or not answering are excluded. The sample includes the 56 treated groups in the analysis sample. Energy-relevant indicates the equipment whose calculated energy use the action can change; (partial) indicates actions that can only do so indirectly. To prevent identification of individual groups, shares are suppressed (–) where 1-4 groups answered "Yes" or 1-4 did not; suppressed shares above 90% are shown as > 90.

Table A.2: Predictors of attrition

Variable	Retained (1)	Attrited (2)	(2)-(1)
Treated	0.514	0.483	-0.031
	(0.502)	(0.509)	
Science Faculty	0.706	0.759	0.052
	(0.458)	(0.435)	
Number of researchers (log)	2.111	2.020	-0.091
	(0.799)	(0.867)	
Baseline annual electricity consumption (MWh)			
Total	22.026	18.678	-3.348
	(36.951)	(36.734)	
Total (log)	2.026	1.489	-0.537
	(1.689)	(1.840)	
Fume cupboards	8.877	1.995	-6.882**
	(28.980)	(4.138)	
Cold storage	6.366	8.100	1.734
	(9.854)	(22.023)	
IT equipment	5.623	4.985	-0.638
	(8.978)	(6.396)	
Other equipment	1.160	3.599	2.438
	(2.259)	(13.050)	
Share any equipment	0.771	0.621	-0.150
	(0.422)	(0.494)	
Share any space	0.679	0.586	-0.093
	(0.469)	(0.501)	
Number of observations	109	29	

Note: This table displays baseline characteristics for groups that were retained in the analysis sample and groups that attrited between baseline and endline.Equipment is grouped into fume cupboards, cold storage (fridges, freezers, and ULT freezers), IT equipment, and other laboratory equipment (cryostats, microbiological safety cabinets, CO_2 incubators, glassware drying cabinets, water baths, and block heaters). Columns (1) and (2) report the mean, with standard deviations in parentheses. Column (2)-(1) reports the difference in means, with significance stars from a two-sided t-test. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

Table A.3: Balance table: pooled sample

Variable	Control (1)	Treatment (2)	(2)-(1)
Science Faculty	0.660	0.750	0.090
	(0.478)	(0.437)	
Number of researchers (log)	2.074	2.146	0.072
	(0.814)	(0.789)	
Baseline annual electricity consumption (MWh)			
Total	20.211	23.744	3.533
	(25.278)	(45.506)	
Total (log)	2.011	2.040	0.030
	(1.725)	(1.669)	
Fume cupboards	8.334	9.391	1.057
	(18.592)	(36.356)	
Cold storage	6.155	6.566	0.411
	(7.531)	(11.703)	
IT equipment	4.533	6.654	2.121
	(6.753)	(10.627)	
Other equipment	1.189	1.133	-0.055
	(2.154)	(2.374)	
Share any equipment	0.887	0.661	-0.226***
	(0.320)	(0.478)	
Share any space	0.679	0.679	-0.001
	(0.471)	(0.471)	
Number of observations	53	56	

Note: This table displays baseline characteristics for the analysis sample, separately for control and treatment groups. Columns (1) and (2) report the mean, with the standard deviation reported in parentheses. Column (2)-(1) reports the difference in means, with significance stars from a two-sided t-test. Equipment is grouped into fume cupboards, cold storage (fridges, freezers, and ULT freezers), IT equipment, and other laboratory equipment (cryostats, microbiological safety cabinets, CO_2 incubators, glassware drying cabinets, water baths, and block heaters). Share any equipment and Share any space report the share of groups that report sharing equipment or space with at least one other group, respectively. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

Table A.4: Energy use by equipment type

	Mean Energy (MWh)	Mean Units	Mean Energy per Unit (MWh)
Overall			
Fume Cupboard	8.877	3.60	2.468
IT Equipment	5.623	14.97	0.376
ULT Freezer	3.113	0.65	4.779
Freezer	1.576	3.30	0.477
Fridge	1.287	3.56	0.361
CO_2 Incubator	0.489	0.89	0.549
Cryostat	0.390	0.08	4.726
Microbiological Safety Cabinet	0.346	0.75	0.460
Glassware Drying Cabinet	0.152	0.20	0.753
Water Bath	0.100	1.26	0.080
Block Heater	0.073	1.78	0.041
Total	22.026	31.05	0.709
Medicine or Joint			
Fume Cupboard	6.964	2.62	2.653
IT Equipment	6.534	20.22	0.323
ULT Freezer	4.667	0.97	4.818
Freezer	3.429	6.81	0.503
Fridge	2.311	6.44	0.359
CO_2 Incubator	1.309	2.41	0.544
Cryostat	–	–	–
Microbiological Safety Cabinet	0.791	1.75	0.452
Glassware Drying Cabinet	–	–	–
Water Bath	0.149	1.84	0.081
Block Heater	0.016	2.09	0.008
Total	26.909	45.50	0.591
Science			
Fume Cupboard	9.672	4.00	2.418
IT Equipment	5.244	12.79	0.410
ULT Freezer	2.467	0.52	4.749
Freezer	0.806	1.84	0.437
Fridge	0.861	2.36	0.364
CO_2 Incubator	0.148	0.26	0.568
Cryostat	–	–	–
Microbiological Safety Cabinet	0.161	0.34	0.478
Glassware Drying Cabinet	–	–	–
Water Bath	0.080	1.01	0.079
Block Heater	0.097	1.65	0.059
Total	19.996	25.04	0.799

Note: This table reports mean baseline annual energy use, mean number of units, and mean annual energy use per unit, by equipment type and faculty. The sample includes the 109 groups in the analysis sample. To prevent identification of individual groups, values are suppressed (–).

Table A.5: Energy-saving tips displayed in calculators

Tip	Equipment	Share of using groups (%)
Don't let that icing get out of hand! If you can de-ice your freezer (once every 6-12 months) you will keep it energy efficient	Freezer	61.3
You're opening the door a lot! Have you used a fridge map or inventory to help you find your stuff? If you can reduce your door openings to 8 per day you can save X kWh per year	Fridge	45.7
This unit is on a lot! If you only used this 255 days per year you would save X kWh per year	CO_2 incubator	>60
You're opening the door a lot! Have you used a freezer map or inventory to help you find your stuff? If you can reduce your door openings to X per day you can save X kWh per year	Freezer	35.5
If you can warm up to -20C you can save X kWh per year	Freezer	29.0
Warming your ULT up to -70 degrees Celsius could save you X kWh per year	ULT freezer	40.9
If you use solid drawers you can save X kWh per year	Freezer	<20
If you can de-ice your freezer (once every 6-12 months) you can save X kWh per year	Freezer	<20
This unit is on a lot! If you only used it for 8 hours a day you would save X kWh per year	Water bath	<25
This unit is on a lot! If you only used this 255 days per year you would save X kWh per year	Water bath	<25
You're opening the door a lot! Have you used a freezer map or inventory to help you find your stuff? If you can reduce your door openings to X per day you can save X kWh per year	ULT freezer	<25
Any tip displayed		50.0
No tips displayed		50.0

Note: This table reports the tips displayed in treated groups' energy use calculators, by equipment type. The sample comprises treated groups in the analysis sample. Share of using groups reports the share of treated groups using that equipment who were displayed each tip. Any tip displayed and No tips displayed report the number of treated groups that received at least one tip and no tips, respectively. To prevent identification of individual groups, shares where 1–4 equipment-owning groups were shown a tip, or 1–4 equipment-owning groups were not, are reported as bounds.

Table A.6: Energy use by equipment type: levels

	Total	Fume cupboard	Fridge	Freezer	ULT	Cryostat	MSC	Incubator	Glassware	Bath	Heater	IT
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Treated × Post	-0.067	-0.018	0.000	-0.050	0.009	—	-0.001	-0.019	0.017	-0.000	0.002	-0.008
	(0.102)	(0.018)	(0.000)	(0.050)	(0.032)		(0.001)	(0.019)	(0.017)	(0.000)	(0.002)	(0.082)
WCB p-value	0.540	0.446	0.698	0.465	0.877		0.456	0.445	0.505	0.460	0.205	0.800
RI p-value	0.507	0.495	0.730	0.277	0.812		0.963	0.223	0.091	0.052	0.356	0.762
Number of observations	218	218	218	218	218	218	218	218	218	218	218	218
Baseline mean	22.026	8.877	1.287	1.576	3.113	0.390	0.346	0.489	0.152	0.100	0.073	5.623

Note: This table reports the coefficient on Treated × Post from separate difference-in-differences regressions of energy use on treatment status, by equipment type, in levels (MWh), estimated using Equation 1. Cryostat is omitted because there is no within-group variation in cryostat energy use over time. Standard errors are clustered at the research group level and reported in parentheses. WCB and RI report wild cluster bootstrap and randomization inference p-values, respectively. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

Table A.7: Energy use by equipment type: logs

	Total	Fume cupboard	Fridge	Freezer	ULT	Cryostat	MSC	Incubator	Glassware	Bath	Heater	IT
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Treated × Post	-0.056	-0.002	0.000	-0.025	0.003	—	-0.000	-0.013	0.004	-0.000	0.002	-0.011
	(0.051)	(0.002)	(0.000)	(0.024)	(0.006)		(0.000)	(0.013)	(0.004)	(0.000)	(0.002)	(0.007)
WCB p-value	0.200	0.446	0.691	0.426	0.626		0.456	0.445	0.505	0.460	0.205	0.086
RI p-value	0.176	0.598	0.885	0.253	0.513		0.725	0.184	0.277	0.083	0.307	0.106
Number of observations	218	218	218	218	218	218	218	218	218	218	218	218
Baseline mean	2.026	1.071	0.579	0.593	0.830	0.118	0.187	0.233	0.076	0.077	0.041	1.357

Note: This table reports the coefficient on Treated × Post from separate difference-in-differences regressions of energy use on treatment status, by equipment type, in logs, estimated using Equation 1. Cryostat is omitted because there is no within-group variation in cryostat energy use over time. Standard errors are clustered at the research group level and reported in parentheses. WCB and RI report wild cluster bootstrap and randomization inference p-values, respectively. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

Table A.8: Treatment effects on energy use: robustness

	Levels				Log			
	Baseline (1)	Robustness (2)	Robustness (3)	Robustness (4)	Baseline (5)	Robustness (6)	Robustness (7)	Robustness (8)
Treated × Post	-0.067	-0.060	0.035	-0.037	-0.056	-0.048	0.001	-0.050
	(0.102)	(0.095)	(0.094)	(0.113)	(0.051)	(0.043)	(0.005)	(0.050)
Number of observations	218	218	218	218	218	218	218	218
Baseline mean	22.026	22.026	22.026	22.026	2.026	2.026	2.026	2.026
Research group FE	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	—	—	—	✓	—	—	—
Faculty × Time FE	—	✓	—	—	—	✓	—	—
Institute × Time FE	—	—	✓	—	—	—	✓	—
Enumerator × Time FE	—	—	—	✓	—	—	—	✓

Note: This table reports the coefficient on Treated × Post from difference-in-differences regressions of energy use on treatment status, in levels (MWh) and logs, under alternative fixed-effects specifications. Column (1)/(5) reproduces the baseline specification from Table 3 (research group and period fixed effects). Columns (2) and (6) replace period fixed effects with faculty × period fixed effects; columns (3) and (7) with institute × period fixed effects; columns (4) and (8) with enumerator × period fixed effects. Standard errors are clustered at the research group level and reported in parentheses. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

Table A.9: Heterogeneity in treatment effects on energy use: levels

	All	Faculty		Baseline Energy Use		Wet or Dry		Group Size	
		Science	Medicine/Joint	Below median	Above median	Dry	Wet	Below median	Above median
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated × Post	-0.067	0.049	-0.323	-0.144	0.008	-0.121	-0.072	-0.175	0.047
	(0.102)	(0.102)	(0.209)	(0.139)	(0.142)	(0.327)	(0.065)	(0.144)	(0.147)
WCB p-value	0.540	0.590	0.068	0.375	0.934	0.699	0.318	0.251	0.731
RI p-value	0.507	0.666	0.019	0.486	0.971	0.506	0.322	0.405	0.731
Number of observations	218	154	64	110	108	64	154	112	106
Baseline mean	22.026	19.996	26.909	3.515	40.879	5.108	29.057	9.310	35.462

Note: This table reports the coefficient on Treated × Post from difference-in-differences regressions of energy use on treatment status, estimated separately by subgroup, in levels (MWh), using Equation 1. Column (1) reports the pooled estimate from Table 3. Dry denotes groups that have only IT equipment; Wet denotes groups that have other lab equipment. Standard errors are clustered at the research group level and reported in parentheses. WCB and RI report wild cluster bootstrap and randomization inference p-values, respectively. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

Table A.10: Heterogeneity in treatment effects on energy use: logs

	All	Faculty		Baseline Energy Use		Wet or Dry		Group Size	
		Science	Medicine/Joint	Below median	Above median	Dry	Wet	Below median	Above median
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated × Post	-0.056	0.002	-0.171	-0.104	-0.003	-0.220	-0.003	-0.110	0.001
	(0.051)	(0.006)	(0.148)	(0.096)	(0.005)	(0.203)	(0.005)	(0.099)	(0.006)
WCB p-value	0.200	0.686	0.209	0.272	0.584	0.281	0.575	0.222	0.906
RI p-value	0.176	0.712	0.400	0.295	0.544	0.467	0.585	0.189	0.965
Number of observations	218	154	64	110	108	64	154	112	106
Baseline mean	2.026	1.938	2.238	0.644	3.433	0.396	2.703	1.190	2.909

Note: This table reports the coefficient on Treated × Post from difference-in-differences regressions of energy use on treatment status, estimated separately by subgroup, in logs, using Equation 1. Column (1) reports the pooled estimate from Table 3. Dry denotes groups that have only IT equipment; Wet denotes groups that have other lab equipment. Standard errors are clustered at the research group level and reported in parentheses. WCB and RI report wild cluster bootstrap and randomization inference p-values, respectively. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

Table A.11: Treatment effects on awareness of energy use

	(1)
Treated	0.388**
	(0.165)
WCB p-value	0.026
RI p-value	0.113
Number of observations	93

Note: This table reports the coefficient on treatment from a regression of awareness of energy and resource costs on treatment status, controlling for log group size, log baseline energy use, and faculty, estimated using Equation 3. The sample comprises 93 groups who answered the awareness and attitudes questions at endline. Awareness is measured on a five-point Likert scale, from 1 (strongly disagree) to 5 (strongly agree). Standard errors are clustered at the enumerator level and reported in parentheses. WCB and RI report wild cluster bootstrap and randomization inference p-values, respectively. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

Table A.12: Changes in reported checklist actions

	All	Not energy-relevant	Energy-relevant	
	(1)	(2)	(3)	(3)-(2)
Share "Yes" at BL (%)	74.3	74.2	74.6	0.4
				(2.2)
Switched to "Yes" (% of pairs not "Yes" at BL)	15.1	13.5	19.0	5.6
				(4.4)
Checklist actions	49	35	14	
Group–action pairs	1446	1033	413	
Group–action pairs not "Yes" at BL	372	267	105	
Switched to "Yes" at EL	56	36	20	

Note: This table reports checklist actions at baseline and endline among treated groups, separately for actions that are and are not energy-relevant. Energy-relevant actions are those that map onto inputs of our equipment energy use measure, either directly (6 actions) or indirectly (8 actions); Appendix Table A.1 gives the classification. The sample includes the 48 treated groups that answered the checklist at both baseline and endline. The unit of observation is a group–action pair, restricted to pairs answered "Yes", "No", or "I don't know" at both baseline and endline; pairs answered "Not applicable" or left unanswered in either period are excluded. A switch to "Yes" is a pair answered "No" or "I don't know" at baseline and "Yes" at endline. Column (3)-(2) reports the difference between energy-relevant and other actions, in percentage points, estimated as the coefficient on an energy-relevant indicator from a linear probability model of the row outcome. Standard errors are clustered at the research group level and reported in parentheses. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

A.2 Figures

Figure A.1: First tab of checklist

SPARK: Sustainable Practices in the Lab checklist

This workbook contains the checklist of actions from SPARK, as well as sheets containing additional information on the actions. The actions are categorized according to certification (bronze/silver/gold).

Please go through the actions and mark which your lab group has already taken. Some of the activities may not be applicable to your lab group - please skip these.

In December 2025, there will be an audit of the actions on this checklist. Completing enough actions from each certification category will result in the award of a certificate.

About SPARK

SPARK is a comprehensive program designed to help laboratories reduce their environmental footprint by improving energy efficiency, water management, waste practices, and procurement. The program provides a structured framework for laboratories to implement sustainable practices, save money, and enhance research quality.

Why should we care about sustainability in the lab?

Laboratories are high-resource environments, consuming up to 10 times more energy than typical office spaces. A single conventional ultra-low temperature (ULT) freezer can use as much energy as an entire house, approximately 20 kWh per day.

Progress towards certification

Certification category	Bronze	Silver	Gold
Number of actions reviewed	8	0	4
Number of actions taken	5	0	2
Progress	63%	0%	50%

Note: This figure shows a screenshot of the first tab of the sustainable practice guidelines checklist given to treated groups, showing introductory information and a summary of an example group's progress towards certification (illustrative responses).

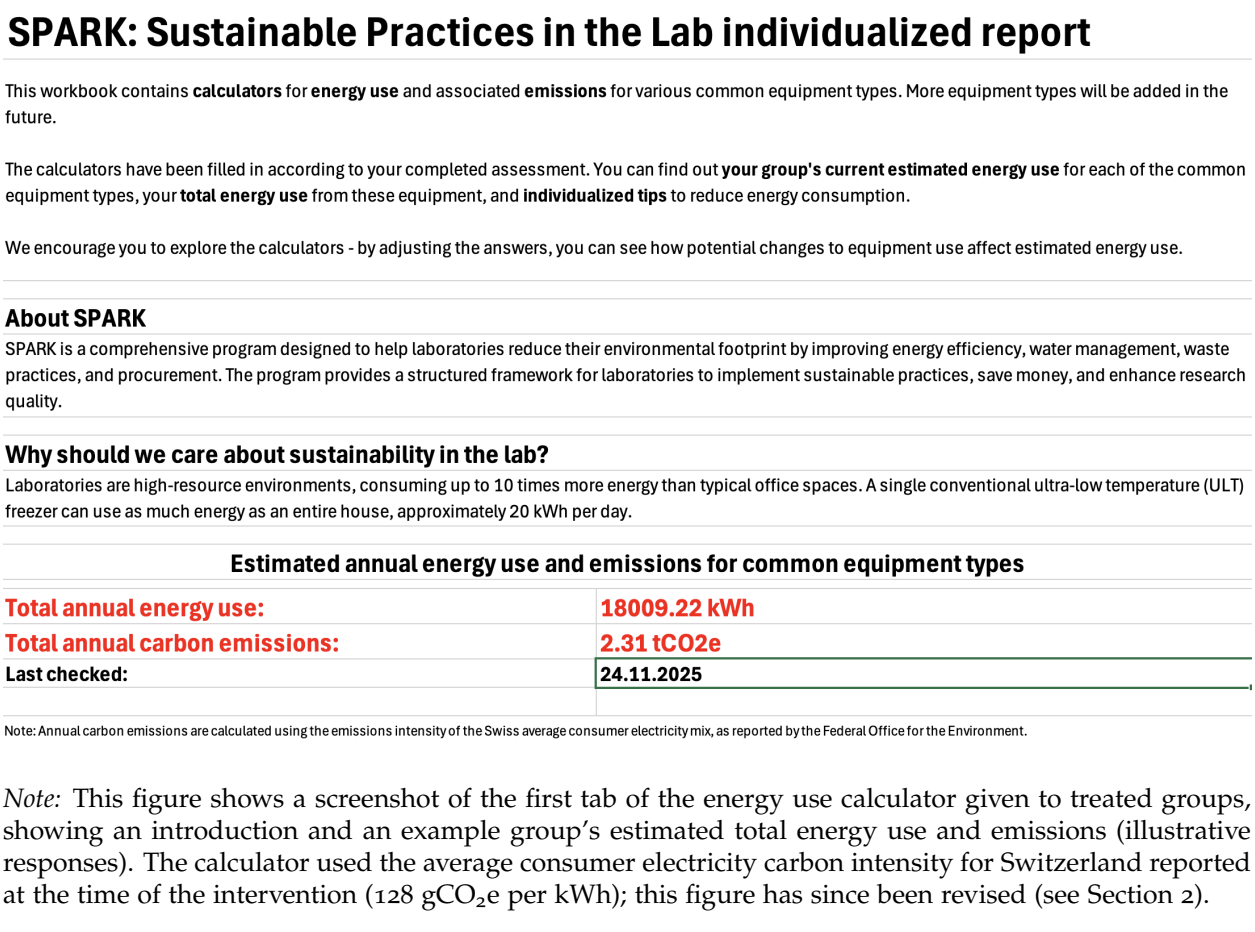
Figure A.2: Bronze checklist



Action	Certification	Has your lab group taken this action?	Why is this important in reducing emissions and costs?	Details
Bronze				
General Lab				
A full assessment of laboratory equipment is completed to determine which items can be switched off and there are systems in place to maintain power management of equipment.	Bronze	I don't know	Laboratory equipment can be energy intensive. Leaving it on can waste energy, as well as increase needs for maintance and risk breakages. Equipment which heats, cools, or moves air should be targeted as these will use the most energy. LCD displays use neglible amounts of energy on their own.	To address this: -Review all equipment in the laboratory -Determine which may be powered down and when -Demark this, be it with stickers, booking systems, etc -Assign who is responsible -Consider implementing digital timers. These can go on piec samples/materials overnight. For e.g. water baths, drying ca There must be evidence in place that equipment has been r can be turned off (unless on timers).
Purified water sources are used appropriately based on an assessment of the purity required for specific tasks.	Bronze	No	Purifying water requires energy and specialised equipment. Avoid excessive use of over-purified water to save energy and materials	Please denote in training or in materials which levels of purif
In-House Production is utilised to reduce wastage and save resources.	Bronze	Yes	In-house production will reduce consumable costs, and the associated packaging. It will also instill a greater understanding of the materials being utilised for experiments	To be considered for in-house production: - Precast gels - Prepared media - Buffers
Recycle: Waste streams are properly separated to/and allow recycling with laboratories	Bronze	No		Packaging materials are collected for recycling rather than d support segregation. To permit recycling, ensure the following has been done: - Recycling bins are made available, and separated from no - A review has taken place of what items may be commonly - Signage is standardised and clear - Training is updated
Onboarding Training: All new laboratory members receive a sustainability	Bronze	Yes	Training is a fanastic way to ensure awareness of good practice is achieved.	This training should cover key topics such as - fume hood operation - freezer management - Reuse schemes

Note: This figure shows a screenshot of some of the bronze-level actions in the sustainable practice guidelines checklist. The responses are ilustrative.

Figure A.3: First tab of calculator



SPARK: Sustainable Practices in the Lab individualized report

This workbook contains **calculators** for **energy use** and associated **emissions** for various common equipment types. More equipment types will be added in the future.

The calculators have been filled in according to your completed assessment. You can find out **your group's current estimated energy use** for each of the common equipment types, your **total energy use** from these equipment, and **individualized tips** to reduce energy consumption.

We encourage you to explore the calculators - by adjusting the answers, you can see how potential changes to equipment use affect estimated energy use.

About SPARK

SPARK is a comprehensive program designed to help laboratories reduce their environmental footprint by improving energy efficiency, water management, waste practices, and procurement. The program provides a structured framework for laboratories to implement sustainable practices, save money, and enhance research quality.

Why should we care about sustainability in the lab?

Laboratories are high-resource environments, consuming up to 10 times more energy than typical office spaces. A single conventional ultra-low temperature (ULT) freezer can use as much energy as an entire house, approximately 20 kWh per day.

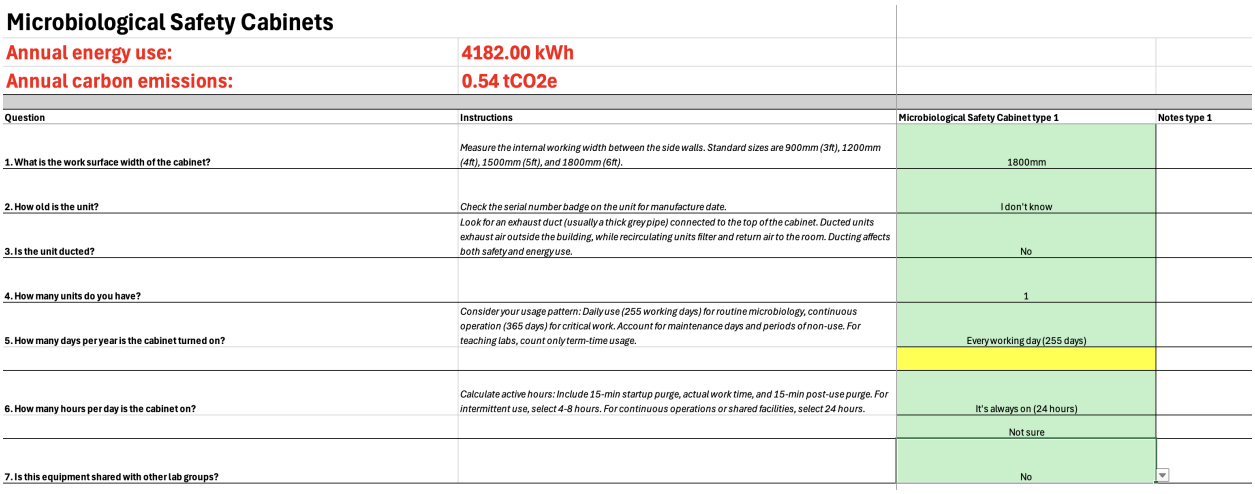
Estimated annual energy use and emissions for common equipment types

Total annual energy use:	**18009.22 kWh**
Total annual carbon emissions:	**2.31 tCO2e**
Last checked:	**24.11.2025**

Note: Annual carbon emissions are calculated using the emissions intensity of the Swiss average consumer electricity mix, as reported by the Federal Office for the Environment.

Note: This figure shows a screenshot of the first tab of the energy use calculator given to treated groups, showing an introduction and an example group's estimated total energy use and emissions (illustrative responses). The calculator used the average consumer electricity carbon intensity for Switzerland reported at the time of the intervention (128 gCO_2e per kWh); this figure has since been revised (see Section 2).

Figure A.4: Microbiological safety cabinets tab of calculator



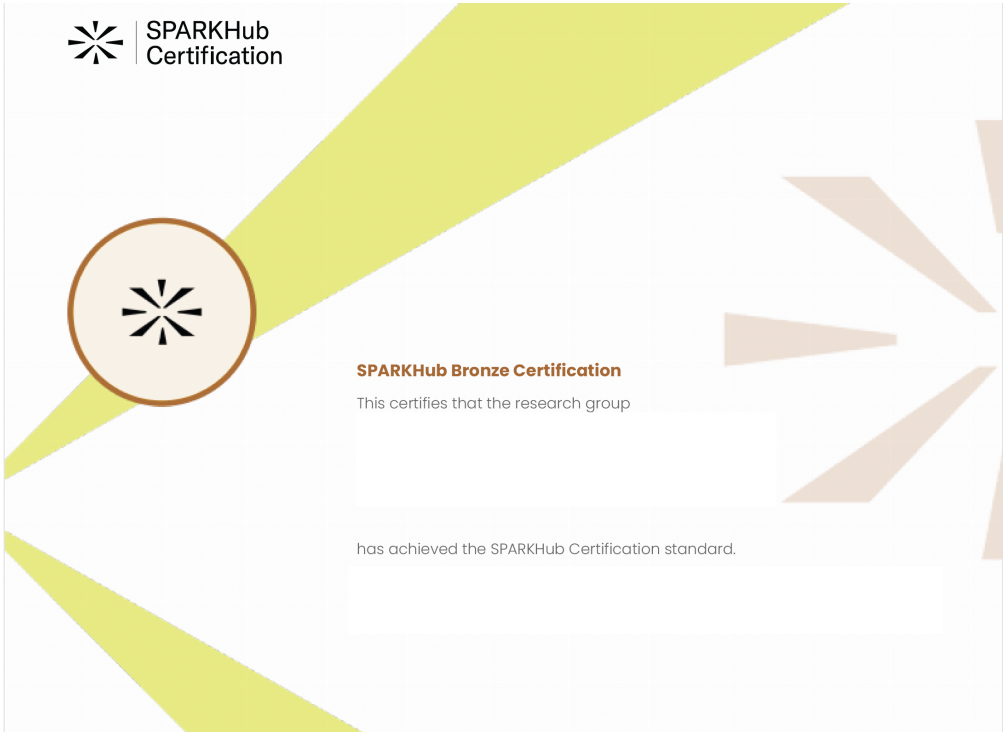
Microbiological Safety Cabinets

Annual energy use:	**4182.00 kWh**
Annual carbon emissions:	**0.54 tCO2e**

Question	Instructions	Microbiological Safety Cabinet type 1	Notes type 1
1. What is the work surface width of the cabinet?	*Measure the internal working width between the side walls. Standard sizes are 900mm (3ft), 1200mm (4ft), 1500mm (5ft), and 1800mm (6ft).*	1800mm	
2. How old is the unit?	*Check the serial number badge on the unit for manufacture date.*	I don't know	
3. Is the unit ducted?	*Look for an exhaust duct (usually a thick grey pipe) connected to the top of the cabinet. Ducted units exhaust air outside the building, while recirculating units filter and return air to the room. Ducting affects both safety and energy use.*	No	
4. How many units do you have?		1	
5. How many days per year is the cabinet turned on?	*Consider your usage pattern: Daily use (255 working days) for routine microbiology, continuous operation (365 days) for critical work. Account for maintenance days and periods of non-use. For teaching labs, count only term-time usage.*	Every working day (255 days)	
6. How many hours per day is the cabinet on?	*Calculate active hours: Include 15-min startup purge, actual work time, and 15-min post-use purge. For intermittent use, select 4-8 hours. For continuous operations or shared facilities, select 24 hours.*	It's always on (24 hours)	
		Not sure	
7. Is this equipment shared with other lab groups?		No	

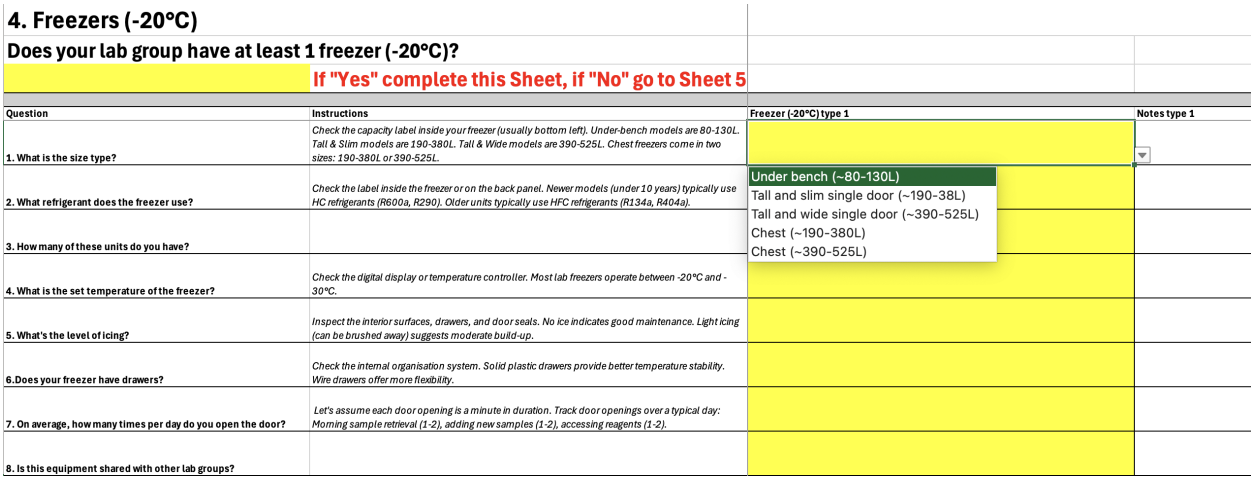
Note: This figure shows a screenshot of the microbiological safety cabinets tab of the energy use calculator, showing the input questions for this equipment type and an example group's responses (illustrative only). The calculator used the average consumer electricity carbon intensity for Switzerland reported at the time of the intervention (128 gCO_2e per kWh); this figure has since been revised (see Section 2).

Figure A.5: Bronze certificate template



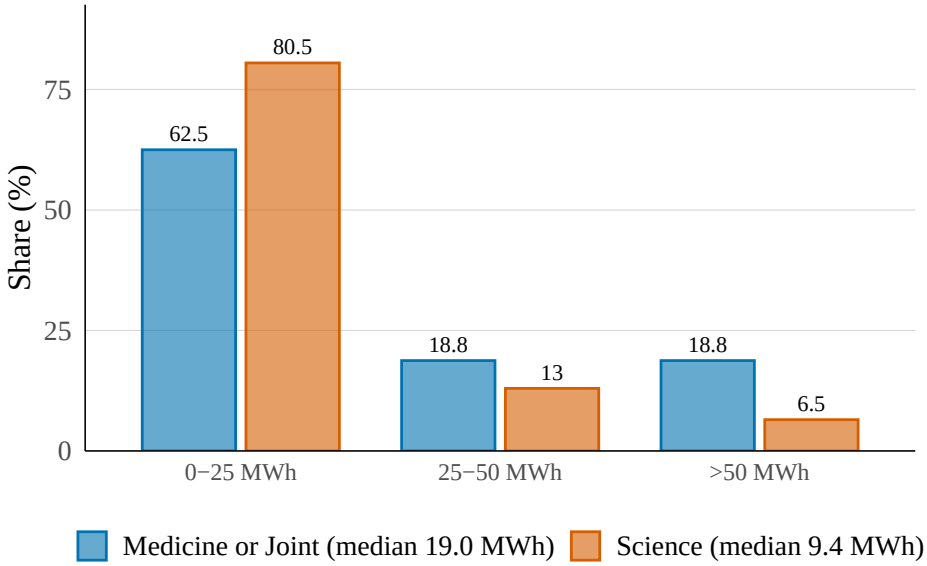
Note: This figure shows the certificate template awarded to treated groups that achieved bronze certification, with group-identifying fields left blank.

Figure A.6: Freezer tab of baseline survey



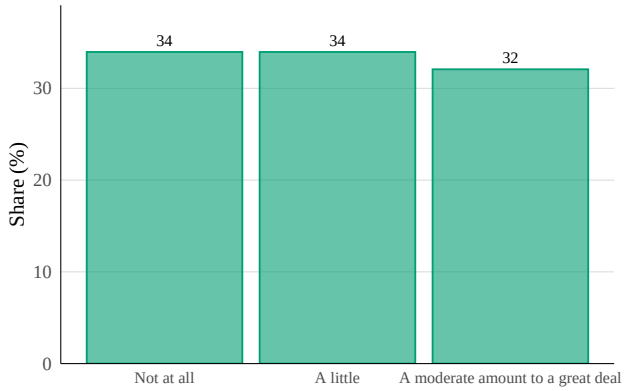
Note: This figure shows a screenshot of the freezers tab of the baseline survey, showing the input questions for this equipment type.

Figure A.7: Annual energy use at baseline by faculty



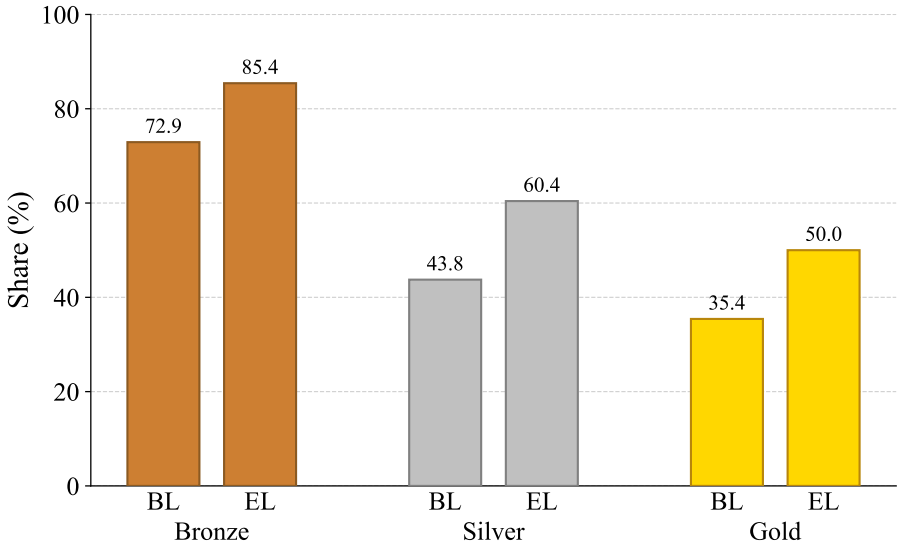
Note: This figure shows the distribution of annual energy use at baseline, separately by faculty, in MWh. The sample includes the 109 groups in the analysis sample.

Figure A.8: Discussion of sustainable practice guidelines checklist



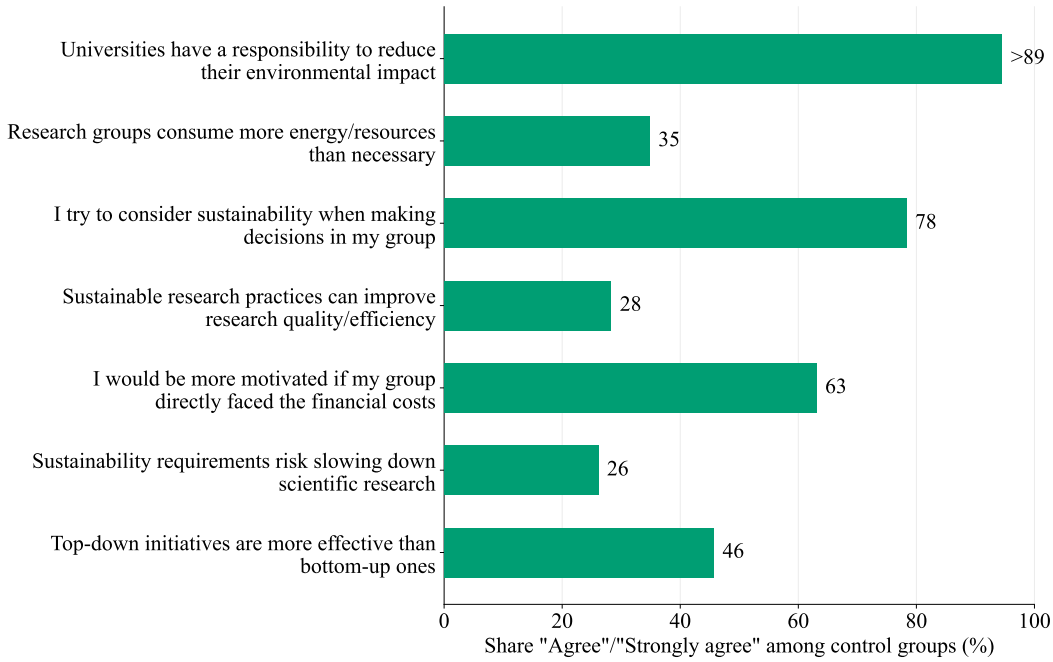
Note: This figure shows the distribution of treated groups" reported discussion of the sustainable practice guidelines between baseline and endline, as reported at endline. The sample includes treated groups in the analysis sample. The answer options "A moderate amount", "Quite a lot", and "A great deal" are combined into one category.

Figure A.9: Certification progress among treated groups



Note: This figure shows the share of treated groups eligible for bronze, silver, and gold certification, at baseline (BL) and endline (EL). The sample includes treated groups in the analysis sample. Baseline eligibility is based on groups' answers to the sustainable practice guidelines checklist at baseline; the certification audit took place at endline only.

Figure A.10: Attitudes among control groups



Note: This figure shows the share of control groups reporting "agree" or "strongly agree" to each attitude statement, at endline. The sample includes control groups in the analysis sample. The exact value for "Universities have a responsibility to reduce their environmental impact" is not reported to prevent identification of individual groups.

B Data Appendix

B.1 Text analysis of publications

Corpus. We combine the title and abstract of each of the 8,014 publications matched to the 95 groups with publications data. We detect the language of each text: 7430 publications (93%) are in English, and of the remaining 584, most are in German (451 publications). We translate the German texts into English using the Helsinki-NLP opus-mt-de-en model. We then represent each publication as a TF-IDF vector of unigrams and bigrams, removing English stop words and terms that appear in fewer than three or more than 95% of publications, yielding 44,350 terms.

Topic model. We fit a non-negative matrix factorization (NMF) topic model to the TF-IDF matrix. We choose the number of topics k by comparing values from 4 to 20 on redundancy between topics, the number of topics dominated by a single group, and reconstruction error, and by inspecting each topic's most heavily weighted terms. We select $k = 7$, the smallest value at which immunology and virology forms a distinct topic. We next exclude three topics for which more than 90% of publications come from a single group, since these reflect individual groups' publication output rather than research areas. Table B.1 lists the most heavily weighted terms for the four remaining topics and the labels we assign to these topics.

Table B.1: Topic labels and most heavily weighted terms

Topic	Most heavily weighted terms
General/broad	species, based, data, high, using
Immunology and virology	cells, cell, immune, virus, human
Collider physics	proton, tev, sqrt, 13 tev, collisions
Flavour physics	pi, mu, pm, decays, decay

Note: This table lists the five most heavily weighted terms for each of the four topics remaining from the NMF topic model with $k = 7$, after excluding three single-group topics, and the labels we assign to them. Terms are unigrams and bigrams after translation and stop word removal.

Group-level topic shares. The topic model assigns each publication a weight on each topic, which we normalize to sum to one. We average these shares across a group's publications and, after excluding the three single-group topics, re-normalize each group's shares over the four remaining topics to sum to one.

Regressions. To understand which topics are associated with higher energy use, we regress log annual energy use on the topic shares, controlling for log research group size,

with and without institute fixed effects. We omit the topic share for the general/broad topic, which serves as the reference category. Table B.2 shows the results.

To examine whether topics can predict energy use beyond administrative characteristics, we compare two LASSO specifications, one without topic shares and one with them. Both include log group size and institute fixed effects, where institutes comprise 12 categories, pooling institutes with fewer than three groups into "other science" and "other non-science" categories. We use LASSO because of the risk of overfitting with 95 groups and up to 15 regressors: LASSO shrinks coefficients and selects regressors, and comparing cross-validated R^2 allows us to test whether research topics add predictive power (Mullainathan and Spiess, 2017). We estimate cross-validated R^2 using nested cross-validation: an outer five-fold split evaluates out-of-sample fit, and an inner five-fold split selects the LASSO penalty, with features standardized within each training fold. Table B.3 shows the results. Cross-validated R^2 is 0.49 (SD across folds 0.04) for the model with administrative characteristics only and 0.61 (SD 0.07) with research topic shares.

Table B.2: Research topics and energy use: OLS estimates

	(1)	(2)
Log(group size)	1.294***	1.190***
	(0.176)	(0.159)
Collider physics	0.229	1.193
	(0.640)	(0.993)
Flavour physics	-0.619	0.057
	(1.100)	(1.271)
Immunology and virology	2.683***	3.545***
	(0.429)	(0.643)
Institute FE		✓
Number of observations	95	95
R^2	0.55	0.72

Note: This table reports OLS estimates from regressions of log annual baseline energy use on log group size and research topic shares, without (column 1) and with (column 2) institute fixed effects. The reference category is the general/broad topic. The sample includes the 95 groups with publications data. Heteroskedasticity-robust standard errors are reported in parentheses. $*p < 0.1, **p < 0.05, ***p < 0.01$.

Table B.3: Determinants of energy use

	Admin only	Admin + research topics
	(1)	(2)
Log(group size)	0.92	0.79
Research topics		
Immunology and virology	—	0.63
Institute FE	✓	✓
Number of observations	95	95
Cross-validated R^2	0.49	0.61

Note: This table reports coefficients from a cross-validated LASSO regression of log annual energy use on log group size, institute fixed effects, and research topic shares. Column (1) includes only administrative characteristics (log group size and institute fixed effects); column (2) adds research topic shares. The sample includes the 95 groups with publications data. Cross-validated R^2 is reported for each specification.

B.2 Surveys

In this section, the questions used in the baseline and endline surveys are provided. All the content below was formatted as an Excel spreadsheet. Unless indicated, all questions appeared in both the baseline and endline surveys for both treated and control groups.

Table B.4: Questions from the characteristics sheet

Question	Instructions	Answer
What is your faculty?		Faculty of Medicine (MeF) / Faculty of Science (MNF)
What is your institute or department?		Free text
How many researchers are in your lab group?		Integer
How many of those researchers work full time?		Integer
How many of those researchers work part time?		Integer
Which rooms does your lab group work in?	Please enter each room on a separate line. Please include the building and floor.	Free text

Table B.5: Questions from the fume cupboards sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 fume cupboard?		Yes / No	If no, entire sheet is skipped
Controller type	The controller will be positioned on the front of the fume cupboard. Often, if it's a VAV controller it will say so. If there's no clue on the controller itself the front panel above the sash may give you a clue. If this panel is solid it's more likely to be a VAV unit, if it's perforated it's more likely to be a CAV controlled unit.	Constant Air Volume (CAV) / Variable Air Volume (VAV)	
Sash width (mm)	Have a ruler or tape measure handy for this. How wide is the aperture at the front?	100 mm increments from 1700 to 2700 / Other	Other for free text option
Design face velocity (m/s)	Affixed to the front of the fume cupboard there should be a card that will state the design face velocity. It'll sometimes have the actual face velocity on there too but don't use that one today.	0.05 m/s increments, 0.35-1.00 / I don't know / Other	Other for free text option
How many of this fume cupboard type do you have?		Integer	
Are all the contents inside your fume cupboard on a plinth, stand or raised tile loading worksurface?	Are your contents lifted up to allow air to flow underneath them?	Yes / No / I don't know	
How many hours per day is the sash open?	If it's every now and then you're looking at 1-3 hours. A busy chemist will have the sash open longer.	Integer, 1-8 / I don't know	Labels to assist, 1 is light usage, 3 is normal, 5 is heavy, 7 is persistent
How many days per year is the sash open?	If it's in use Monday to Friday each working week select 255 days.	365 / 255 / I don't know / Other	Other for free text option
Please estimate how much of your worksurface is occupied by your equipment and apparatus.	No need to be exact, just enter how much of the surface area you think is covered.	0-100%, 10% increments	
Is this equipment shared with other lab groups?		Yes / No	
Compared to October, how often is the sash open?	Please indicate how the time the sash is open has changed compared to when the assessment was completed in October.	Less often / About the same / More often	**Only on endline survey**

Table B.6: Questions from the fridges sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 fridge (4°C)?		Yes / No	If no, entire sheet is skipped
What is the size type?	Check the capacity label inside your fridge (usually bottom left). Under-bench models are typically 100-160L. Tall single-door models range from 250-700L. Double-door models are 1200-1400L.	Under bench convection (∼ 100-160L) / Under bench fan assisted (∼ 100-160L) / Tall single door fan assisted (∼ 250-425L) / Tall single door fan assisted (∼ 500-700L) / Tall double door fan assisted (∼ 1200-1500L)	
On average, how many times per day would you estimate the door to be opened?	Consider your lab's routine: Morning check (1), sample storage/retrieval (2-4), reagent access (2-3), end-of-day storage (1).	Integer	
How many of these units do you have?		Integer	
Is this equipment shared with other lab groups?		Yes / No	
Compared to October, how often is the door opened per day?	Please indicate how the number of daily door openings has changed compared to when the assessment was completed in October.	Less often / About the same / More often	**Only on endline survey**

Table B.7: Questions from the freezers sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 freezer (-20°C)		Yes / No	If no, entire sheet is skipped
What is the size type?	Check the capacity label inside your freezer (usually bottom left). Under-bench models are 80-130L. Tall & Slim models are 190-380L. Tall & Wide models are 390-525L. Chest freezers come in two sizes: 190-380L or 390-525L.	Under bench (∼ 80-130L) / Tall and slim single door (∼ 190-380L) / Tall and wide single door (∼ 390-525L) / Chest (∼ 190-380L) / Chest (∼ 390-525L)	
What refrigerant does the freezer use?	Check the label inside the freezer or on the back panel. Newer models (under 10 years) typically use HC refrigerants (R600a, R290). Older units typically use HFC refrigerants (R134a, R404a).	HFCs (usually an older unit) / HCs (usually a newer unit	
How many of these units do you have?		Integer	
What is the set temperature of the freezer?	Check the digital display or temperature controller. Most lab freezers operate between -20°C and -30°C.	-20°C to -30°C by one degree increments / I have no way of seeing this	
What's the level of icing?	Inspect the interior surfaces, drawers, and door seals. No ice indicates good maintenance. Light icing (can be brushed away) suggests moderate build-up.	None / A little building on the drawers and shelves / Yes it's 2mm thick and can make opening the door/accessing contents challenging	
Does your freezer have drawers?	Check the internal organisation system. Solid plastic drawers provide better temperature stability. Wire drawers offer more flexibility.	Yes they are solid plastic / Yes they are made from plastic coated wire / No there are no drawers or most are missing	
On average, how many times per day do you open the door?	Let's assume each door opening is a minute in duration. Track door openings over a typical day: Morning sample retrieval (1-2), adding new samples (1-2), accessing reagents (1-2).	Integer	
Is this equipment shared with other lab groups?		Yes / No	
Compared to October, how often is the door opened per day?	Please indicate how the number of daily door openings has changed compared to when the assessment was completed in October.	Less often / About the same / More often	**Only on endline survey**

Table B.8: Questions from the ULT freezers sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 ULT freezer?		Yes / No	If no, entire sheet is skipped
What type of unit is your ULT freezer?	Check the model information on the front or side panel. Stirling engine models are distinctive and newer. For compressor types, check the age and look for refrigerant info (HC = hydrocarbon, HFC = hydrofluorocarbon). Newer models (under 10 years) typically use HC refrigerants.	Stirling engine with hydrocarbon refrigerants / Cascade compressors with hydrocarbon refrigerants, under 10 years old / Cascade compressors with hydrocarbon refrigerants, over 10 years old / Cascade compressors with hydrofluorocarbon refrigerants / Dual compressor with hydrofluorocarbon refrigerants, under 10 years old / Dual compressor with hydrofluorocarbon refrigerants, over 10 years old / Dual compressor with hydrocarbon refrigerants / I don't know	
What is the size type?	Check the capacity label inside the freezer (usually bottom left). Most ULT freezers are tall, single-door units. Measure internal volume if unsure: 500-600L is standard, 700-800L is large, and 800-900L is extra large.	Small single door (¡500L) / Tall single door (∼500-600L) / Tall single door (∼700-800L) / Tall single door (∼800-900L)	
What is the set temperature of the freezer?	Check the digital display for the current set temperature. Most ULTs operate at -80°C for long-term biological sample storage. Some applications use -70°C or -75°C.	-70°C / -75°C / -80°C	
How many of these units do you have?		Integer	
The door seals are?	Inspect all door seals carefully. Clean seals with no visible damage ensure optimal performance. Dirty/iced seals indicate maintenance needs. Damaged/missing seals severely impact efficiency and should be replaced immediately.	Clean and in good condition / Dirty or lightly iced / Damaged, absent in places, or with a layer of ice	
Spacing, my ULT freezers have...	Measure the clearance around your ULT. Proper spacing (≥150mm) ensures efficient heat dissipation. Items stored on top block ventilation and force the compressor to work harder.	≥150mm gap at the back and sides of the unit, with no items stored on top / There's little space around the unit and we store items on top of the unit.	
What is the condition of the filter?	Locate the condenser filter (usually behind front grille). A clear filter allows proper airflow. Dirty filters restrict airflow and reduce efficiency. No filter or severely clogged filters can lead to compressor failure.	No filter / It's clean / It's a little dirty / Most of it is clogged	
On average, how many times per day do you open the door?	Let's assume each door opening is a minute in duration? Monitor door openings: Morning checks (1), sample storage/retrieval (2-3), end-of-day additions (1). Each minute the door is open significantly impacts temperature recovery.	Integer	
Is this equipment shared with other lab groups?		Yes / No	
Since October, has the set temperature of the freezer changed?	Please indicate whether the set temperature of the freezer has changed since the assessment was completed in October.	Yes the set temperature has increased / Yes the set temperature has decreased / No the set temperature has not changed	**Only on endline survey**

Table B.9: Questions from the glassware drying cabinets sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 glassware drying cabinet?		Yes/No	If no, entire sheet is skipped
What is the capacity of the drying cabinet?		∼90-100L / ∼180-200L / ∼425-550L / ∼850-1000L	
Which statement best describes the cabinet?		The chamber is insulated, it's warm to the touch when on (modern technology) / The chamber has no insulation, it's hot to touch when on (older technology)	
Does the chamber use a fan to circulate air?		Yes / No / I don't know	
What is the set temperature?		50°C / 60°C / 70°C / I don't know	
How many units do you have?		Integer	
How many days per year is the drying cabinet turned on?		365 / 255 / Other	Other for free text option
How many hours per day is the drying cabinet turned on?		24 / Other	Other for free text optoin
Is this equipment shared with other lab groups?		Yes / No	
Compared to October, how often is the cabinet turned on?	Please indicate how the cabinet is turned on compared to when the assessment was completed in October.	Less often / About the same / More often	**Only on endline survey**

Table B.10: Questions from the microbiological safety cabinets sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 microbiological safety cabinet?		Yes/No	If no, entire sheet is skipped
What is the work surface width of the cabinet?	Measure the internal working width between the side walls. Standard sizes are 900mm (3ft), 1200mm (4ft), 1500mm (5ft), and 1800mm (6ft).	900mm / 1200mm / 1500mm / 1800mm	
How old is the unit?	Check the serial number badge on the unit for manufacture date.	Less than 5 years old, 5-20 years old / More than 20 years old / I don't know	
Is the unit ducted?	Look for an exhaust duct (usually a thick grey pipe) connected to the top of the cabinet. Ducted units exhaust air outside the building, while recirculating units filter and return air to the room. Ducting affects both safety and energy use.	Yes / No	
How many units do you have?		Integer	
How many days per year is the cabinet turned on?	Consider your usage pattern: Daily use (255 working days) for routine microbiology, continuous operation (365 days) for critical work. Account for maintenance days and periods of non-use. For teaching labs, count only term-time usage.	365 / 255 / I don't know / Other	Other for free text option
How many hours per day is the cabinet on?	Calculate active hours: Include 15-min startup purge, actual work time, and 15-min post-use purge. For intermittent use, select 4-8 hours. For continuous operations or shared facilities, select 24 hours.	24 / Other	Other for free text option
Is this equipment shared with other lab groups?		Yes / No	
Compared to October, how often is the cabinet turned on?	Please indicate how the cabinet is turned on compared to when the assessment was completed in October.	Less often / About the same / More often	**Only on endline survey**

Table B.11: Questions from the cryostats sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 cryostat?		Yes / No	If no, entire sheet is skipped
What is the chamber set temperature?	Check the digital display for the current set temperature. Most cryostats operate between -18°C and -25°C depending on the tissue type and cutting requirements.	-18°C / -20°C / -25°C	
Does your cryostat have a sleep mode or energy saving mode?	This mode will warm the chamber temperature to -10°C to save energy. Check your cryostat's manual or settings menu for an 'ECO mode', 'Sleep mode' or 'Energy saving mode'. This feature is common in newer models.	No, it does not / Yes, but it's not used / Yes, we use the unit 9am to 5pm, outside these hours it's in sleep mode / Yes, we use the unit 8am to 5pm, outside these hours it's in sleep mode / Yes, we use the unit 8am to 7pm, outside these hours it's in sleep mode	
How many units do you have?		Integer	
How many days per year is the cryostat heater turned on?	Consider your sectioning schedule. For routine histology work select 255 days (weekdays). For facilities with variable demand, count the actual days of use. Some labs keep units running continuously (365 days) due to long stabilization times.	365 / 255 / I don't know / Other	Other for free text option
How many hours per day is the cryostat turned on?	For routine sectioning work (9-5), select 8-9 hours. If you have multiple users throughout the day, select longer hours. Consider that cryostats need time to stabilise at operating temperature.	24 / I don't know / Other	Other for free text option
Is this equipment shared with other lab groups?		Yes / No	
Since October, have the sleep mode settings changed?	Please indicate whether the sleep mode settings have since the assessment was completed in October.	Yes it is in sleep mode more often / Yes it is in sleep mode less often / No the sleep mode settings have not changed	**Only on endline survey**

Table B.12: Questions from the water baths sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 water bath?		Yes / No	If no, entire sheet is skipped
What is the capacity of the water baths?	Check the capacity label or measure the tank volume. Small baths (5-6L) are typically for test tubes/small vessels. Medium baths (10-14L) handle multiple containers. Large baths (18-20L) are for high-throughput work.	5-6L / 10-12L / 10-14L / 18-20L	
What is the bath using to heat contents?	NaN	Water / Metal Beads	
What is the set temperature?	Check the digital display or dial setting. Common settings: 37°C for cell culture, 65°C for enzyme reactions, 90°C for special applications.	37°C / 65°C / 90°C / Other	Other for free text option
How many units do you have?		Integer	
Does your water bath have the lid on?	Check if appropriate lids are available and used.	Yes / No	
How many days per year is the water bath turned on?	Consider your usage pattern: Daily use (255 working days), continuous operation (365 days), or occasional use. Account for holidays and maintenance periods. For shared facilities, check booking systems or logs.	365 / 255 / I don't know / Other	Other for free text option
How many hours per day is the water bath turned on?	Calculate active hours: Include warm-up time (30-60 mins) plus actual usage. For intermittent use, consider timer controls. 24-hour operation is common for shared facilities.	24 / I don't know / Other	Other for free text option
Is this equipment shared with other lab groups?		Yes / No	
Compared to October, how often is the water bath turned on?	Please indicate how often the water bath is turned on compared to when the assessment was completed in October	Less often / About the same / More often	**Only on endline survey**

Table B.13: Questions from the CO_2 incubator sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 CO_2 incubator?		Yes / No	If no, entire sheet is skipped
What is the capacity of the incubator?	Check the capacity label or measure the internal volume. Small units (150-170L) are suitable for personal/small lab use. Medium units (220-260L) are for shared facilities.	∼150-170L / ∼220-260L	
What is the age of the unit?	Check the serial number or documentation for manufacture date. Units $\leq$15 years typically have better insulation and CO_2 sensors.	$\leq$20 years / > 20 years	
How many units do you have?		Integer	
How many days per year is the incubator turned on?	Most cell culture facilities operate continuously (365 days) due to long-term experiments and the need to maintain stable conditions. For teaching labs or seasonal work, count only active usage days plus stabilization periods.	365 / 255 / I don't know / Other	Other for free text option
Is this equipment shared with other lab groups?		Yes / No	
Compared to October, how often is the incubator turned on?	Please indicate how often the incubator is turned on compared to when the assessment was completed in October	Less often / About the same / More often	**Only on endline survey**

Table B.14: Questions from the block heaters sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 block heater?		Yes / No	If no, entire sheet is skipped
How many blocks does this unit heat as standard?	Look at the number of block positions on your heater. Most lab block heaters have 2-4 block positions.	1-Block / 2-Block / 3-Block / 4-Block	
What is the set temperature?	Check the digital display or dial setting. Common temperatures are 37°C for biological samples, 65°C for enzyme reactions, and 95-100°C for DNA denaturation.	37°C / 65°C / 95°C / 100°C / Other	Other for free text option
How many units do you have?		Integer	
How many days per year is the block heater turned on?	Consider your typical usage pattern. For routine lab work select 255 days (weekdays). For continuous operations or cell culture work, select 365 days.	365 / 255 / I don't know / Other	Other for free text option
How many hours per day is the block heater turned on?	Think about your typical protocol times. For short incubations select 1-3 hours, for day-long protocols select 8-12 hours, for continuous operations select 24 hours.	24 / I don't know / Other	Other for free text option
Is this equipment shared with other lab groups?		Yes / No	
Compared to October, how often is the block heater turned on?	Please indicate how often the block heater is turned on compared to when the assessment was completed in October	Less often / About the same / More often	**Only on endline survey**

Table B.15: Questions from the IT equipment sheet

Question	Instructions	Answer	Notes
Does your lab group have at least 1 IT equipment?		Yes / No	If no, entire sheet is skipped
Equipment type?	Check your computer specifications. Standard laptops/desktops are for basic office/lab work. High-powered units with dedicated graphics cards are for data analysis or computational work.	Standard laptop / High-powered laptop with graphics card / Standard desktop computer / High-powered desktop computer with graphics card	
How many of these units do you have?		Integer	
How many days per year is the IT kit in use?	Consider your work schedule: Standard year (255 working days), continuous operation (365 days), or part-time use. Account for holidays and remote work days.	365 / 255 / I don't know / Other	Other for free text option
How many hours per day is your IT equipment on?	Calculate active hours including startup time. For standard workday, select 8-10 hours. For extended use, select actual hours.	24 / I don't know / Other	Other for free text option
What external monitor does your computer use?	Measure screen size diagonally. Consider brightness settings: Low (25-30%), Mid (50-60%), Full (100%).	Small (19-22 inches), lowest brightness / Small (19-22 inches), mid brightness / Small (19-22 inches), full brightness / Medium (24-27 inches), lowest brightness / Medium (24-27 inches), mid brightness / Medium (24-27 inches), full brightness / Large (30+ inches), lowest brightness / Large (30+ inches), mid brightness / Large (30+ inches), full brightness	
How many screens do you have?	Count all active displays connected to your computer.	1 / 2 / 3 / 4 / 5	
Is this equipment shared with other lab groups?		Yes / No	
Compared to October, how often is the IT kit turned on?	Please indicate how often the IT kit is turned on compared to when the assessment was completed in October	Less often / About the same / More often	**Only on endline survey**

Table B.16: Questions from the waste sheet

Question	Answer	Notes
What is your weight of Recycling waste/week	0-3kg / 4-6kg / 7-9kg / 10-12kg / Other / I don't know	Other for free text option
What is your weight of Clinical waste/week	0-3kg / 4-6kg / 7-9kg / 10-12kg / Other / I don't know	Other for free text option
What is your weight of General waste/week	0-3kg / 4-6kg / 7-9kg / 10-12kg / Other / I don't know	Other for free text option
Notes on waste type	Free text	Available for each type of waste

Table B.17: Questions from the specialized equipment sheet for the following machines: PCR machine, Ice machine, Centrifuge, Coffee machine, Microwave Animal facility, Non-CO_2 incubator, 4°C room, -20°C room, and Other (write in notes)

Question	Answer	Notes
Does your lab group have this equipment?	Yes / No	
If yes, how many units?	Integer between 1-10	
If yes, is it shared with another lab group?	Yes / No	
Notes on equipment	Free text	Available for each piece of equipment

Table B.18: Questions from the sharing and communications sheet

Question	Instructions	Answer
Do you share any equipment with other lab groups?	Write "Yes" if previously answered that share any of the equipment in this survey with other lab groups or if share any other equipment not asked about in this survey with other lab groups.	Yes / No
Which lab groups do you share equipment with?	Please enter each lab group on a separate line. If answer "No" to 1, leave blank.	Free text
Do you share lab space with any other lab groups?		Yes / No
Which lab groups do you share lab space with?	Please enter each lab group on a separate line. If answer "No" to 3, leave blank.	Free text
How often do you share lab space with other lab groups?	How often is another lab group working in the same lab space as you? If answer "No" to 3, leave blank.	Always / Frequently / Rarely / We never use the same lab space at the same time
Communication with other lab groups: Which lab groups do you talk with?	Can be talking about research or informally. Please enter each lab group on a separate line. Please indicate how often you talk to each of these lab groups.	Free text
Communication with other lab groups: How often do you talk with each of these lab groups?	Can be talking about research or informally. Please enter each lab group on a separate line. Please indicate how often you talk to each of these lab groups.	Daily / Weekly / Monthly / Less than monthly

Table B.19: Questions from the consent sheet

Question	Answer
We want to better understand research networks between laboratories at the University of Zurich. For this purpose, we ask your permission to collect publicly available information on research collaboration and co-authorship between you and other researchers at the university who also give us permission, and to merge it with your survey data. You may withdraw your consent for this data collection and merging at any time.	Yes I consent to this data collection and merging / No I do not consent

Table B.20: Questions from the awareness sheet

Question	Instructions	Answer	Notes
Since October, how much have you discussed the SPARK checklist within your research group?		Not at all / A little / A moderate amount / Quite a lot / A great deal	**Only for treated groups at the endline**
Have you heard of any lab sustainability programs such as SPARK or LEAF?		Yes / No / Not sure	**Only for control groups at the endline**
When did you first hear about this?	Only if answer "Yes" to the previous question	Within the last 2 months / 2-6 months ago / 6-12 months ago / More than 1 year ago	**Only for control groups at the endline**

Table B.21: Questions from the climate attitudes sheet

Question	Answer
The next part of the survey contains questions on your climate attitudes and pro-environmental behaviors. Participation in this part of the study is optional. You may decline to answer these questions without affecting your participation in the main study or how your other data are used. You may withdraw your consent to participate in this part of the study at any time. Do you consent to participate in this part of the study?	Yes I consent to participate in this part of the study / No I do not consent
Universities have a responsibility to reduce their environmental impact.	Strongly disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree
Research groups consume more energy and resources than necessary for their scientific work.	Strongly disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree
I try to consider sustainability when making decisions about equipment, protocols, or practices in my research group.	Strongly disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree
Sustainability requirements risk slowing down scientific research.	Strongly disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree
Sustainable research practices can improve research quality or efficiency.	Strongly disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree
I am generally aware of the energy and resource costs associated with my research group's activities.	Strongly disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree
I would be more motivated to adopt energy-saving practices if my research group directly faced the financial costs of its energy and resource use.	Strongly disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree
Initiatives from the university administration are more effective in changing research culture than those from within research groups.	Strongly disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree